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RESEARCH ARTICLE

Robust Visual Servoing and CNN-Based Thermal Imaging for Sugarcane Row Following With a Skid-Steering Mobile Robot

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ABSTRACT The escalating demand for accuracy and efficiency in agriculture—driven by population growth and climate change—has increased the use of mobile robots in precision agriculture. Sugarcane production is particularly significant, providing both sugar yield and renewable energy sources such as bioplastic and biofuel from cane bagasse. Like other grass-type crops (e.g., wheat, corn, rice, barley, and bamboo), sugarcane fields pose unique challenges due to vast plantation areas and dense vegetation forming hard-to-access crop row tunnels. These conditions hinder both manual labor and autonomous robot operation, requiring higher investments in robot swarms with robust navigation systems to handle noisy imaging and low Global Navigation Satellite System coverage. This paper introduces a visual navigation system for TIBA, a cost-effective, compact, rugged skid-steering mobile robot equipped with customized sensors and designed for swarm-based operations in sugarcane fields. We propose a combination of infrared (thermal) imaging with robust control to address the locomotion challenge in sugarcane fields. Thermal images reveal temperature differences between ground and cane stalks, enabling clearer visualization of the row corridor despite occluding leaves. Consequently, computationally efficient algorithms, such as a simple convolutional neural network, can generate reliable trajectories for the robot to follow along the crop row, avoiding lateral collisions with the side plants. A sliding mode control approach ensures robust tracking under uncertainties and disturbances in both robot and camera models. We also introduce IRNavSim, a Matlab-based simulation environment tailored for crop row following tasks using thermal imaging, which helps us to evaluate the system's performance and behavior.

INDEX TERMS Precision agriculture, wheeled mobile robots, autonomous crop row following, robust control.

I. INTRODUCTION

In recent years, robotic systems applied to precision agriculture have been a rising research area in the automation field due to their great potential to obtain increased food production, enhanced precision, and safety of crop operations [1]. These assets have been brought to light due to growing concern about climate change, a more sustainable environment, and increased demand for food [2].

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The growth of the world's population has encouraged farmers to seek an increase in the efficiency of food production, for example, by reducing the waste of consumables and increasing the productivity of small areas of cultivation at the lowest possible cost. The design of robots for agriculture represents a particular challenge for research, since each plantation field is characterized by a completely different environment, harvesting procedure, safety issues, and many tasks to be performed, which must be taken into account in the development of the mechanical design and navigation system. Among basic agricultural tasks, we can highlight

land plowing, fertilizing, sowing, and irrigating, while the complex tasks consist of precision agriculture, including fruit harvesting, weeding, pruning, and plant phenotyping. There is currently in the world a wide variety of mobile robots deployed for agricultural purposes, which illustrates how robotic technologies are mature and incorporated into agricultural systems, as well as their great innovation potential for the next decades [3]. These vehicles are usually equipped with specialized tools, multiple sensors, and robot manipulators to perform several types of agricultural tasks. In this direction, we can highlight, for example, the following cutting-edge technologies: Thorvald II, a modular and re-configurable multitask platform, carrying UV-light devices for the treatment of mildew and fungus in strawberry polytunnels, in Norway, and grapevine crops in open fields, in France and the United States [4]; Delta-type robot arms for precision spraying in vineyards (Ecorobotix SA), in Switzerland, and non-chemical weeding in cereal crops (Small Robot Co.), in the United Kingdom. As mentioned above, these requirements do not exclusively concern Europe and North America, but also many countries in Central and South America. In the specific case of Brazil—a country of continental size—the agricultural industry is already experiencing a high degree of automation for the planting and harvesting of soybeans, cotton, coffee, rice, corn, wheat, and sugar cane, mainly in large areas, with only a few cases of robotics-based applications. More recently, sustainable target-spray applications have been deployed using the Solix scouting robot (Solinftec Inc.) in soybean, corn, wheat, and cotton crops, as well as the Arbus 4000 JAV spraying robot (Jacto Inc.) in citrus farms. Yet, this high degree of automation is not a reality for local farmers and growers to carry out agricultural tasks in small areas (e.g., vegetable gardens and orchards).

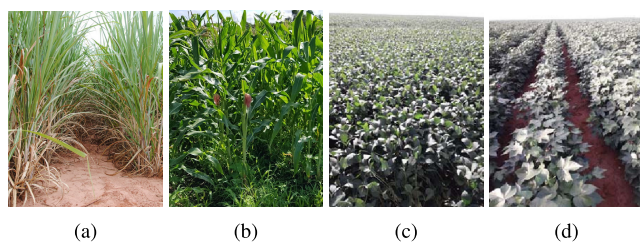


FIGURE 1. Grass-type crops in Brazil - (a) sugarcane, at Umoe BioEnergy production field, Presidente Prudente-SP; (b) corn, at Embrapa Agrobiologia, Seropédica-RJ. Short-plant crops in Brazil northeastern region- (c) soybeans; (d) cotton. Images from authors' database.

In recent years, sugarcane has been a focus of great interest, not only for sugar production but also for environmentally friendly biofuel, pulp and paper from the cane bagasse [5]. As an essential alternative to fossil fuels, it can be used as a clean and renewable source of energy, allowing for the reduction of petroleum consumption and greenhouse gas emissions and stimulating the generation of other valuable products (e.g., bio-plastics and bio-electricity). Farms for sugarcane production usually demand gigantic areas for

planting, which requires high investment in hiring many employees and machinery, as well as increased costs for logistics and maintenance. In addition, most of the tasks carried out on sugarcane fields are laborious and tedious, susceptible to human errors and performed in a harsh environment subjected to bad weather conditions (e.g., extreme temperatures and thunderstorms), dense vegetation, and wildlife. Recently, robotic vehicles for sugarcane crops have been the subject of intense research, as well as for other grass-type crops in general, such as wheat, rice, barley, bamboo and corn, where we can highlight the Rowbot, a mobile vehicle for monitoring and applying in-season nitrogen to corn crops [6], and TIBA, a skid-steering mobile robot for agricultural tasks in sugarcane fields [7]. Examples of existing work in applications carried out for sugarcane production include: harvesting efficiency [8], weed detection and control [9], mapping of misplanting and skips [10], analysis of billet quality [11], and fertilization [12]. One of the major challenges encountered in sugarcane crops is to develop a robust scheme for robot locomotion—as well as for other grass-type crops in general shown in Fig. 1a and 1b—due to limited GNSS signal obstructed by dense and tall foliage that may interfere with effective navigation, and due to the formed tight crop tunnels that highly restrict the robot design dimensions. Although several solutions for the navigation problem are fairly established for small open fields [13], structured indoor facilities (e.g., greenhouses, nurseries and polytunnels [14]) and open fields with defined seeding pattern and low height plants (e.g., soybeans and cotton), as shown in Fig. 1c and 1d, the research still counts with notable open points that warrant further investigation for poorly structured, GNSS-denied environments with tall plants.

In addition to these navigation challenges, most of the existing robot perception solutions employ laser scanners, LiDARs, and GNSS/IMU [15] for accurate robot mapping—especially Real Time Kinematics (RTK-GNSS) [16], recognized for its precision of around 2 cm and growing popularity in precision agriculture—which can be expensive and unsuitable for cost-sensitive designs, as discussed in [17]. Pioneering research with SLAM (Simultaneous Localization and Mapping) and sensor fusion have been conducted for many navigation problems found in agriculture. For example, we highlight the autonomous navigation system of the BoniRob—a phenotyping robot from Bosch—which utilizes data from 3D MEMS LIDAR and IMU, and the dynamic obstacle detection approach for agricultural robots based on multi-sensor information fusion [18]. On the other hand, as sugarcane fields are gigantic and poorly structured, they may require a swarm approach to cover the entire application area—demanding the robot unit cost to drop—or higher-resolution (more expensive) sensors. As sugarcane fields normally lie in wide areas subjected to extreme sunlight incidence, sun sensors have the potential to assist robot navigation, as these devices are low-cost, use simple technology, and are easy-to-install. They are usually applied

for tracking sunlight inclination, having been already used, for example, in visual odometry update and navigation of planetary rovers [19]. Recent studies have also addressed the development of low-cost navigation systems for GNSS-denied environments—e.g., RoboNav [20], which is also a cost-effective solution, but combines a dual GNSS setup with multiple inertial sensors and a Gaussian Sum Filter to achieve precise and robust heading and position estimation, validated in vineyard rows using an autonomous tracked robot.

In the context of sugarcane environments, infrared (IR) technology emerges as a low-cost alternative solution with the potential to improve row/lane recognition beyond the capabilities of conventional RGB cameras. IR imaging has also been applied in various robotic navigation applications in recent decades. The use of IR cameras for navigation is driven by the capacity of passive IR imaging technology to identify objects under limited visibility conditions with minimal attenuation, such as nighttime, light dust, smoke, and haze [21]. By detecting thermal radiation naturally emitted from objects, it avoids the need for active energy transmission sources, thereby mitigating risks, such as those posed by laser-based mapping. In recent years, IR imaging technology has become more affordable and portable to be mounted on small robotic platforms. For example, IR sensors have been employed in UAV navigation due to GNSS denial, size, and weight constraints faced by this vehicle type [22]. Another interesting IR use case includes determining environment feature temperature and distance to the robot as polynomial functions of the measured IR intensities [23]. Recent works have also explored low-cost solutions, for example, testing IR sensors integrated with RGB-D cameras for in-field 3D geo-referenced mapping and environmental modeling in a commercial vineyard [24].

Within the IR camera market, it is worth noting Teledyne FLIR[®] Systems, Seek Thermal[®], Thermoteknix[®], and Raytheon[®] as manufacturers that offer a broad spectrum of models for a wide range of applications and budgets, being FLIR[®] often regarded as the industry leader. We can highlight the entry-level series FLIR[®] One as a fairly established low-cost option with a modest frame rate of 8.7 Hz and pixel resolution of 120×160 suitable for consumer applications and compatible with USB-connection for smartphones. In comparison, over the past decade, we have witnessed the emergence of higher-end models, such as: (i) FLIR[®] Lepton Series, notable to be a more affordable high-end micro-camera with higher frame rate of 30 Hz, easily embeddable in small mobile robots and drones; (ii) FLIR[®] Machine Vision lineup (e.g., Dragonfly S, Forge, Firefly, Blackfly, Oryx, and Flea), as high-cost solutions for complex robotic systems, industrial inspection, and urban surveying, engineered with more advanced sensors (e.g., Dragonfly is up to 49 Hz and pixel resolution of 2592×1944) and capabilities for both automated and precise manual image control. However, it is worth highlighting that importing/exporting IR cameras are tightly regulated around the world due to their dual-use capability for both

civilian and military applications, raising national security, non-proliferation, and privacy concerns. These regulations are rooted in decades of evolving international treaties and national laws, such as the Wassenaar Arrangement, ITAR (International Traffic in Arms Regulations), and restrict camera specifications like resolution and frame rate to prevent misuse. Hence, cameras exceeding certain performance thresholds may require special licensing due to their potential for military application (FLIR[®] One and Lepton are considered ITAR-free).

In image processing, it is worth mentioning several established techniques and applications for collecting and analyzing field data applied to digital agriculture and generating reference features for navigation and control of mobile robots. In this context, artificial intelligence (AI) has emerged as a cutting-edge technology applied not only to leverage computer vision for agricultural robots [25], but also to enhance general precision agriculture tasks such as plant monitoring, cultivation, and harvesting. In data collection and analysis for agriculture, we can highlight the use of deep learning approaches for weed classification for plant-specific treatment [26], early detection of disease in sugarcane plants [27], monitoring and classification of crop type, health, and maturity using vegetation indexes [28].

Particularly for mobile robot navigation, supervised deep learning using convolutional neural networks (CNNs) for image semantic segmentation has enormous potential to overcome several undesired situations encountered in the field that hinder image processing for precise feature detection, such as blurred image, low image contrast, and especially for sugarcane crops, objects (i.e., leaves and stalks) occluding camera lens [29]. For example, we can mention the use of deep learning-based semantic image segmentation for crop row guidance using adaptive multi-ROI in a strawberry field [30], robust vision-based crop row detection in a sugar beet field [31], visual autonomous navigation in row-based crops with synthetic data generation [32], and study of the effects of dataset diversity on the accuracy and robustness of CNN models for crop row detection [33]. Deep learning combined with IR imaging has also already been addressed in agriculture for several applications such as crop water stress assessment [34], wildlife recognition [35], pathogen detection [36], and fruit yield estimation [37]. To our knowledge, the use of a thermal imaging-based deep learning approach for robot navigation in sugarcane fields was first addressed in [38].

Lane detection has also gained traction in contemporary research, particularly in the domain of self-driving cars, as evidenced by [39]. Several works have successfully integrated CNN architectures with Hough transforms to discern road stripes and margins [40]. While deep learning is typically associated with high computational demands, leading to high hardware costs, this primarily applies to the training phase. Once the CNN model is trained, the processing requirements for real-time inference in the crop field are generally manageable. In addition, computational

demand may be less restrictive depending on navigation requirements because the system can cope with minor discrepancies, for example, in scenarios where high accuracy is not paramount and robot operations expect slow navigation speeds. This flexibility is due to the robot's ability to react against potential inaccuracies in image processing by self-maneuvering/adjusting in real time with a minor risk of obstacle collisions.

Several mobile robot configurations have been extensively studied and explored in response to the varied and challenging terrains encountered in agricultural fields, including sugarcane crops. Wheeled mobile robots have been a primary focus due to their efficiency and versatility in different driving profiles [41]. Differential-drive vehicles consist of two independently driven wheels—usually adding a caster for stability—offering improved precise control and design compactness. Four-wheel drive vehicles, also called all-wheel drive or 4WD, provide independent traction to all four wheels, enhancing stability and control on rough and uneven terrains. Omnidirectional mobile robots, with four-wheel driving (4WD) and four-wheel steering (4WS) configuration, are capable of driving in any direction in a 2D plane and providing better maneuverability. A skid-steering mobile robot (SSMR) [42] delivers the same traction for all wheels (two or more) on the same side of the vehicle. For an SSMR vehicle, curved maneuvers and in-place rotations are achieved by applying different speeds to the left and right wheels, which also forces skidding over the driving surface. Consequently, this creates a challenging requirement for the operation of an SSMR in agricultural environments: low friction between the wheels and the driving terrain. It is essential to highlight that SSMRs are nonholonomic systems, that is, reaching arbitrary poses in the plane will depend on the path taken to achieve it [41], [42]. In practice, unlike differential-drive vehicles, the kinematic modeling of SSMRs may lead to a less precise odometry model, although solutions for this problem have been recently addressed in literature, such as slip prediction from friction-based kinematic model [43]. On the other hand, SSMRs have attracted significant attention among all-wheeled configurations due to their low-cost appeal (i.e., mechanical simplicity), the remarkable agility and applicability for all-terrain purposes, differently from differential-drive configurations. In this context, several robotics companies have already developed commercial solutions for different applications, such as Thorvald (Saga Robotics) for mildew control in strawberry fields using UV-C light treatment; RB-Vogui (Robotnik®), Husky and Jackal (Clearpath Robotics) serving as general purpose and customized platforms for outdoor tasks and field validations for R&D in agriculture, construction and logistics.

While classic control theory may struggle to cope with nonlinearities, disturbances, and unmodeled dynamics, several approaches based on adaptive and robust control have already been addressed. Among relevant research, we can highlight the established approach from Kozłowski and Pazderski that uses time-varying kinematic oscillator

applied for SSMRs based on nonholonomic integrator [42], adaptive steering control for a farm tractor with varying yaw rate [44], nonlinear model predictive control [45], and PID controller tuned by a genetic algorithm applied for surveillance SSMRs with robustness to terrain-dependent values (i.e., rolling resistance, friction and shear deformation modulus) [46]. In the last decade, the sliding mode control (SMC) [47] has been more intensely utilized, as it offers robustness to a variety of nonlinear systems with respect to parametric uncertainties and external perturbations, for example, a sliding mode variable structure for path following control [48] and a second-order terminal SMC for trajectory tracking [49]. While conventional first-order SMCs are subject to the chattering effect, second-order SMCs, specifically the super-twisting algorithm (STA) [47], aim to attenuate chattering by ensuring finite-time convergence without the drawbacks of the terms based on the classical sign function. As mentioned above, the motion control performance for several wheeled robot types, such as DDMRs and SSMRs, are inherently affected by nonholonomic constraints [50]. In this context, coordinate and input transformations grounded in Lie algebra are used to convert nonholonomic systems and their constraints into a more manageable representation for the SMC design, as it translates robot kinematics into a chained form using trigonometric functions [51]. Examples of recent research include super-twisting SMC for skid-steering robots [52] and adaptive multiple-surface SMC for nonholonomic systems with uncertainties [53]. The control scenario also includes Position-Based Visual Servoing (PBVS), Image-Based Visual Servoing (IBVS) and Hybrid Visual Servoing (HVS) approaches. The main difference among them is related to the coordinate system from which the feedback is calculated. While in the PBVS method, feedback is computed using reconstructed Euclidean information to estimate the target's three-dimensional (3D) Cartesian pose concerning the camera pose, the IBVS method computes the feedback directly from extracted features in the image space. The HVS method was proposed to benefit from the advantages of IBVS and PBVS approaches while avoiding the disadvantages of both [54]. Following this trend, unlike [52], [53] that do not consider use of camera feedback, we can highlight the work from [55], which combines SMC and STA approaches with IBVS, using column (IBVS-CC) and row controllers (IBVS-RC) for autonomous crop row following using differential-drive vehicles.

To address the challenges and opportunities discussed above, we propose a robust visual servoing and CNN-based thermal imaging for sugarcane row following with a skid-steering mobile robot, named TIBA - Tankette for Intelligent BioEnergy Agriculture. The main contributions of this work are summarized as follows: (i) deployment of TIBA robot, a compact and cost-effective mobile platform for swarm-based operation in wide sugarcane crops, with some field experiments assessment; (ii) proposal of a novel solution for autonomous crop row-following tasks using a PBVS methodology, which combines IR imaging and SMC

approach to address the challenges posed by grass-type plantation environments; (iii) development of IRNavSim, a simulation environment in Matlab specifically designed for crop row following using IR imaging. This work is structured as follows: Section II provides an overview of TIBA's design, concept and model, findings from the analysis of field-collected data and preliminary experiments, as well as revisits the IR imaging based CNN approach for detecting the robot's offset relative to the crop row centerline, according to [38]; Section III delves into the robust Sliding Mode Control (SMC) strategy for precise crop row tracking, coping with disturbances posed by terrain, modeling inaccuracies, and odometry/IR camera feedback; Section IV introduces the novel Matlab-based simulation environment (IRNavSim) for sugarcane crops utilizing thermal imaging; Section V discusses the simulation results; and Section VI concludes with final thoughts and directions for future research. The paper includes an accompanying video clip,¹ with complementary images and videos related to this work. This manuscript is an extended and more detailed version of the authors' previous publications [7] and [38], which focus respectively on TIBA robot construction aspects and the CNN-based detection of sugarcane crop rows centerlines. In contrast to [38], here we include the robust control design and analysis, additional simulation/experimental results, and the simulation environment (IRNavSim) to emulate a typical sugarcane scenario for autonomous navigation based on thermal images.

II. TIBA - TANKETTE FOR INTELLIGENT BIOENERGY AGRICULTURE

In this section, we briefly present a mobile robotics technology named TIBA (Fig. 2), a skid-steering mobile robot (SSMR) developed by the authors and intended to carry out monitoring tasks in sugarcane fields [7]. From the perspective of construction cost and environmental protection, TIBA was developed with a skid-steering driving profile, utilizing only two motors, and designed to be compact, user-friendly, dust and waterproof, as well as adaptable for the integration of various sensing systems, depending on the agricultural tasks that are required. Consequently, the control system meant for TIBA also prioritizes minimizing overall costs such as electronic components, energy consumption, and algorithm complexity. We also introduce a control scheme that integrates IR thermal imaging technology with the SMC approach into a stand-alone PBVS methodology to address the mobility challenge in sugarcane fields with irregular topography and subject to interference from grasses, stones, and holes. The thermal images captured in the sugarcane environment uncovered significant temperature disparities between the soil and plants or stalks, which allows for a much better distinguishment of the feature that represents the row corridor despite numerous cane leaves obstructing the visual

field. This thermal effect motivates using computationally efficient CNNs to segment the corridor feature in the image effectively. Consequently, this extracted feature generates a reliable trajectory for the robot to follow along the crop, maintaining its position centered between the row cane stalks and updating its local odometry-based navigation system.

In the current development and experimental stages targeted for the sugarcane production crop of Umoe BioEnergy—a Norwegian company located in Presidente Prudente, State of São Paulo, Brazil (Fig. 2a and 2b)—TIBA is envisioned for field mapping, plant cataloging, sampling of cane stalk and soil for *in loco* or *online* analysis, detection of misplanting (i.e., empty planting spots), and management of insect pest and disease. Beyond its primary functionalities, TIBA is also envisioned for future developments to serve as strategic landing platforms for partner drones (Fig. 2c), aiming at symbiotic operation between the two autonomous systems: battery recharging and data storage. While the drones would carry out aerial mapping of the sugarcane plantation for overhead plant health identification, TIBA would focus on tasks within the crop row tunnels, carrying out similar plant health identification missions, but from beneath the foliage coverage, providing more comprehensive assessments of the entire crop. Additionally, TIBA's top lid can be adaptable for solar panel mounting, offering a sustainable energy source to increase the robot's autonomy and serving as a charging station for the partnered drones, providing an integrated solution for more advanced agricultural operations. As aforementioned, the ground navigation problem for this robot along crop rows is especially tricky not only due to the environmental and cost limitations but also due to immature analysis of skid effects due to the mandatory slippage phenomenon for this driving profile, despite existing relevant research on motion control and path planning of nonholonomic systems, as previously mentioned.

The conceptual design of TIBA robot (Fig. 3), including its driving profile and mechanics, embedded electronics, and software architecture, has already been detailed in the previous work of the authors [7]. It was devised based on the constraints and challenges encountered in grass-type environments, which are more accentuated in sugarcane crops. As mentioned above, the Umoe BioEnergy production field (Fig. 4) was used as a relevant case study environment for this particular setup due to its vast covered area and production yield. According to Umoe BioEnergy website,² its field has a crushing capacity of more than 2.5 million tons of sugarcane bagasse on 40,000 ha for ethanol production primarily (170,000 m³).

The main constraints imposed by this environment are the robot's width and height due to the tight and dense plant tunnels formed by the canes, whose leaves lean or fall towards the row centre and may form obstacles for locomotion (even for the farm personnel). The seeding points

¹Check the accompanying video clip versions also at <https://youtu.be/sz7cq4BDtJM> (long) or <https://youtu.be/qR1QuiquKq4> (short).

²Umoe Bioenergy - Website: <http://umoebioenergy.com>, last access: 2025-07-09.



FIGURE 2. TIBA robot motion scenes. Remotely controlled operation in Umoe BioEnergy production field, Presidente Prudente-SP, Brazil - (a) crop row infiltration; (b) robot motion along crop row with multi-sensing system for data collection. Autonomous navigation in Embrapa Agrobiology experimental field for control validation, Seropédica-RJ, Brazil - (c) drone aerial view.

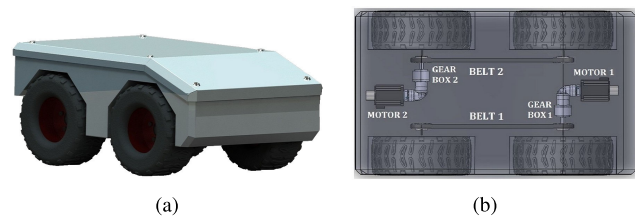


FIGURE 3. TIBA mechanical system - (a) simple conceptual sketch; (b) skid-steering mechanical arrangement with two motors, gear boxes and belts.

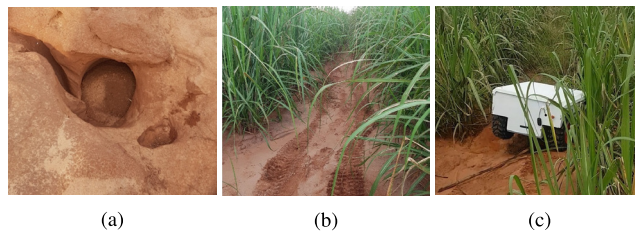


FIGURE 4. Field images at Umoe BioEnergy farm, Presidente Prudente-SP, Brazil - (a) crevices; (b-c) uneven terrain, mud and clay.

of crop rows at Umoe BioEnergy farm—which form nearly straight lanes—are preset by GNSS-given coordinate, and cleared by approximately 0.40 m along the row and by 1.50 m across the lane, having each seeding point also a set of seeds planted around it by 0.25 m. A robot width of 0.72 m for the first prototype was considered enough to allow the vehicle to be driven along the lanes without colliding with the side cane stalks. The wheels were embedded entirely within the robot's maximum width for a more compact solution. The robot height was set to 0.53 m, which was considered enough to allow for a suitable field-of-view (FoV) for any sensors and cameras to be mounted on the robot top and also to avoid trapping leaves to get stuck in the robot structure. Embedding the wheels into the robot chassis was also a solution for dealing with the trapping leaves, which decreases the probability of the branches getting stuck around the wheels' rims and shafts. Other constraints for the driving profile include the crop terrain, which is primarily rough, uneven and full of crevices due to frequent rain action. The soil type is Oxisol, which has a homogeneous texture with clay contents in the range of 20 to 30% and sand in the range of 50 to 70%. This requires the tire design

to cope with friction specifications and also contributes to the formation of crevices. Finally, the extreme weather (i.e., frequent storms, flooding, and temperatures above 40°C during summer) and the wildlife found in this field require a chassis design to be sealed against dust, dripping water and shallow immersions for puddles, as well as the specification of electronic components into a broader range of operating temperature and heat sink.

To the best of the authors' knowledge, TIBA is the first prototype of a four-wheeled mobile robot created for carrying out precision agricultural tasks in sugarcane plantations. Similar robotic systems have been employed in vineyards, orchards, and corn production in the United States, such as the Rowbot system.³ Driven by this motivation, this work builds on the authors' previous work [7], which detailed TIBA's design, construction, and preliminary sensor data collected from field. In the current study, we present new field test results with a focus on the analysis of IR data collected during these experiments. The paper also delves into the proposed strategy for TIBA's row-following within crop tunnels, utilizing IR imaging and a robust control approach based on sliding modes. Although overall cost reduction remains the primary objective of TIBA's navigation system design, the integration of IR imaging with data from other sensors (such as LiDAR, sun sensors, and RGB cameras) could be investigated in future work to enhance system accuracy and reliability.

A. ROBOT MODELING

In this work, we consider the sugarcane field to be the same as that found in Umoe BioEnergy farm. The sugarcane field can be modeled as two rows of cane stalks separated by $d_{row} = 1.50$ m clearance approximately, and being the corridor (the crop row) formed between the plants an almost straight lane along with the mobile robot shall drive without colliding with the side cane stalks (Fig. 5a). For this environment, we consider the ground as a locally flat surface and a kinematic model of the SSMR based on the kinematic modeling approach introduced by [42]. Here, as shown in Fig. 5b, we consider a land vehicle with 3-DoF described

³ROWBOT in the Field - Website: <https://www.rowbot.com/blog-posts/2014/9/8/rowbot-in-the-field>, last access: 2025-07-09.

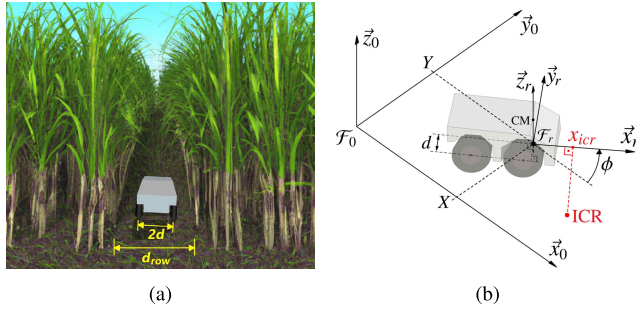


FIGURE 5. TIBA modeling - (a) sugarcane crop row; (b) robot coordinates, parameters, CM and ICR.

by a vector of generalized coordinates $q = [x \ y \ \phi]^T$ that describes the configuration of the robot frame $\mathcal{F}_r$ with respect to the inertial frame $\mathcal{F}_0$. The robot frame origin is assigned to the vertical projection of the robot center-of-mass (CM) on the ground. In this work, we consider the CM to lie in the midpoint of the robot width, but not necessarily at the midpoint between the two wheels along the robot length. Vector $\vec{x}_r$ points towards the robot front, and $\vec{z}_r$ is normal to the robot chassis. The distance between the center of any wheel to the CM point along $\vec{y}_r$ is defined by d . It is well known from the literature that SSMRs have nonholonomic—or non-integrable—kinematic constraints [51]. In addition, SSMRs introduce additional challenges for modeling and control design [42] because of the slippage phenomenon—always present for this configuration—which determines a non-unique and time-varying *instant center of rotation* (ICR) for the mobile robot. Finally, the ICR depends on the desired drift and may lead to instabilities if not properly designed.

It is well-known that a *kinematic* control approach can ensure satisfactory performance when the robot motion demands low speeds and slow accelerations [51]. However, if these conditions are not satisfied, a dynamic control approach is required. According to the kinematic framework, we consider that the mobile robot has a low-level control loop that imposes any specified velocities on the robot wheels, say:

$$\eta = [v_x \ \omega]^T = r \begin{bmatrix} \frac{\omega_L + \omega_R}{2} & \frac{-\omega_L + \omega_R}{2d} \end{bmatrix}^T, \quad (1)$$

where r is the wheel radius (assumed to be the same for all wheels) and ω_L, ω_R are the angular velocities of the left and right wheels, respectively. Notice that $\omega_1 = \omega_2 = \omega_L$ and $\omega_3 = \omega_4 = \omega_R$, being the wheels numbered clockwise from the back-left wheel. The following kinematic map describes the robot motion:

$$\dot{q} = S(q)\eta, \quad S(q) = \begin{bmatrix} \cos(\phi) & x_{icer} \sin(\phi) \\ \sin(\phi) & -x_{icer} \cos(\phi) \\ 0 & 1 \end{bmatrix}, \quad (2)$$

where x_{icer} is the position of the robot ICR along $\vec{x}_r$ -axis, which is arbitrarily defined. The nonholonomic constraint is written in the Pfaffian form as $A(q)\dot{q} = 0$, where $A(q) = [-\sin(\phi) \ \cos(\phi) \ x_{icer}]$.

B. DATA COLLECTION AND PRELIMINARY EXPERIMENTS

This section presents the experimental verification and validation tests performed with TIBA robot in different sugarcane environments and preliminary discussions, interpretations and motivations around the field collected data.

1) VERIFICATION AND DATA COLLECTION IN PRODUCTION FIELD

TIBA robot prototype was taken to Umoe BioEnergy production field—in Presidente Prudente, State of São Paulo, Brazil, as described in Subsection II—aiming for field verification of the requirements for the developed mechanical concept, for understanding the sugarcane environment through data collection from different sensors, and for investigation of possible solutions for robot navigation in such a complex environment, seeking for future enhancements. The sensors employed for the field experiment included RGB and infrared (IR) thermal cameras, LiDAR sensors, a humidity/temperature (HT) sensor, and a sun sensor (Fig. 6). Data from sun and HT sensors were acquired through an external Arduino interface. Data acquisition was carried out in different rounds and recorded in .bag files by using the ROS framework. The primary focus of this research is on the images collected from the IR cameras, while the analysis of data from the LiDAR, HT, and Sun Sensor, as outlined in previous work [7], is beyond the scope of this study.

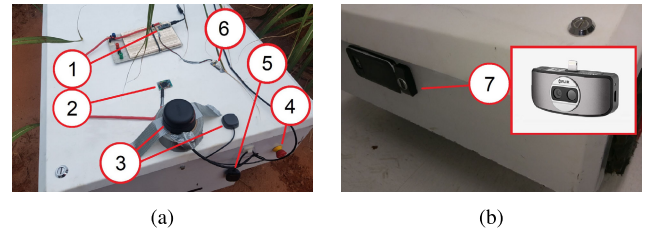


FIGURE 6. Field test preparation - External sensors mounted on robot lid: (a) [1]-Arduino, [2]-Sun sensor, [3]-LiDAR sensors, [4]-E-Stop button, [5]-Joystick antenna, [6]-HT sensor; (b) [7]-IR/RGB camera mounted on the chassis front.

2) IN-SITU MECHANICAL TESTS

TIBA robot was driven through a remote control along several lanes of Umoe BioEnergy production field to evaluate its mechanic structure and gain knowledge about the sugarcane crop. In this context, the robot was subjected to operational trials along various crop rows within the sugarcane field showing satisfactory performance. Remarkably, the morphology of these rows was characterized by smooth and extensive curvatures (Fig. 7a and 7b).

From satellite imagery analysis, we can observe that the “sharpest” corridors display radius of curvature of approximately 125 m (Fig. 7c), while the majority of the rows displayed even more smoothness, with radius of curvature of approximately 1000 m (Fig. 7d), highlighting the absence of tight turns or pronounced bends in the terrain under study.

The designed boxed-type structure proved to be robust and compliant for operations in uneven terrain, with notable



FIGURE 7. Field images at Umoe BioEnergy farm, Presidente Prudente-SP, Brazil. Regions of interest with test crop rows infiltrated by TIBA robot (satellite view adapted from Google Maps database, around geographic coordinates 22.6 deg South, 51.8 deg West) - (a-b) General aerial views; (c) sharper rows with shorter radius of curvature ($R \approx 125$ m); (d) smoother rows with larger radius of curvature ($R \approx 1000$ m). Distance scales indicated at bottom.

performance when passing over crevices (already illustrated in Fig. 4) formed by rainwater accumulation. The selected skid-steering scheme was revealed to be suitable for the desired motion profile, allowing for forward/backward motion, rotation around the longitudinal axis, and curves by differential wheel speeds. During the robot infiltrations along the tunnels formed by the sugarcane stalks and plants (Fig. 8), we assessed the robot adaptability to canes with varying stalk heights: approximately 1, 2, and 3 m. For cane tunnels with 1 to 2 m height, the robot was found seamlessly maneuvering for forward motion, encountering no obstructions. For 3 m or taller cane tunnels, the elongated plant leaves extending inward towards the row centerline introduced soft barriers. These crop characteristics offered reduced operational clearance between the robot flanks and the adjacent stalks, raising the potential for collisions. Nevertheless, there were no instances of leaves becoming stuck in the robot framework or wheels/rims, which underscored the efficacy of its compact design paradigm.



FIGURE 8. Field images at Umoe BioEnergy farm, Presidente Prudente-SP, Brazil - (a) crop row tunnels; (b) TIBA being driven along crop rows tunnels during a field test.

3) RGB AND IR THERMAL CAMERAS

As aforementioned, a FLIR[®] One V2, containing a stereo set of one RGB camera and one infrared (IR) thermal sensor, was attached to TIBA robot's hood for capturing images and videos of the sugarcane crop rows. The main motivation was to understand which features we could extract from the environment under the plant coverage of the row crop tunnel since the trapping leaves offer a very noisy scenario that may complicate a fair detection of the corridor centerline that the robot must continue following during its movement. For the sugarcane field context, the IR thermal camera appears as an alternative solution to better detect the crop rows' soil in addition to the common visual information of a regular RGB camera. During the field tests with TIBA robot at Umoe BioEnergy farm, most of the image collection was performed

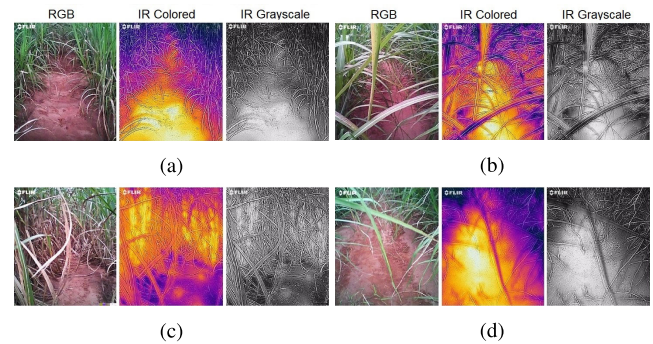


FIGURE 9. Thermal image comparison. Examples of RGB and IR images (false colors and grayscale) from four different places/situations along the crop rows. During a sunny afternoon, around 3:00 PM - (a) 1 m canes; (b) 2 m canes; (c) 3 m canes. Under cloudy weather in the late afternoon, after 6:00 PM - (d) 2 m canes.

during sunlight and a late afternoon, under sunny and cloudy conditions. Notice in Fig. 9 that, compared to regular RGB pictures, the IR images provide a much more distinguishable shape of the crop corridor ground along which the robot must ride, due to the thermal difference between soil and plants.

From the analyzed data, we can observe that the corridor path ahead remains distinguishable even if cane leaves are occluding the camera sensor, regardless of the weather and daytime. For shorter cane stalks (1 to 2 m, Fig. 9a and 9b), the ground appears much warmer than the plants, and a cold “kerb” can be noticed at the lane border, indicating that the base of the seeding point is even colder, as expected from water accumulation. In the case where the leaves are very tall (3 m or more, Fig. 9c), the ground temperature showed to be colder than the plants with no visible kerb, which is arguably due to the less amount of sunlight able to reach the land. The images for a cloudy afternoon (Fig. 9d) presented no changes compared to an early sunny afternoon, which indicates that the thermal pattern can be held even when sunlight fades out, which is due to the soil's high thermal capacity. Due to logistics limitations, we did not consider this experiment to extend through nighttime. The mapping of a complete thermal profile for different day times, locations, year seasons, cane heights and field types, aiming for a complete understanding of the sugarcane soil's thermal capacities and limits, is planned for future research. For example, farmers usually clean the crop row soil in production fields, while experimental fields may have leaves covering the soil, which negatively impacts the thermal image, as further outlined in Subsection II-B4.

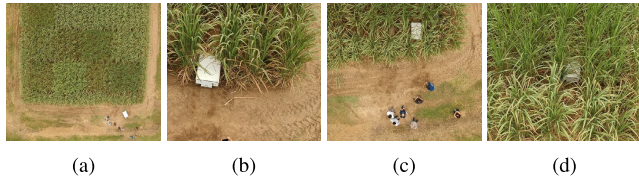


FIGURE 10. Aerial photos of a field test at Embrapa Agrobiology experimental field, Seropédica-RJ, Brazil. TIBA manual and automatic control experiments within the sugarcane area - (a) Phase A, driving to row entrance; (b) Phase B, aligning to crop row; (c-d) Phase C, driving along crop row.

4) VALIDATION IN EXPERIMENTAL FIELDS

New field tests were carried out at the Embrapa Agrobiology experimental field—in Seropédica, state of Rio de Janeiro, Brazil (Fig. 10)—an area owned by the Brazilian Agricultural Research Corporation (Embrapa) with several experimental crops for study in sugarcane, corn and forage grass. These experiments were designed mainly to validate the adaptability of the robotic system to different sugarcane crops and the automatic control for pose regulation. It is worth mentioning that experimental fields such as Embrapa Agrobiology offer distinct characteristics to production crops, which plays a critical role in interpreting field results. It can be observed that the experimental fields feature narrower and shorter row lengths and present more uniform soil topology when compared to production fields with uneven terrains because experimental fields are concentrated in smaller controlled areas. The soil of this experimental field is predominantly of Planosol type, which has a coarser texture than Oxisol, presenting around 10 to 15% clay and more than 80% sand. Moreover, it was observed that the experimental fields are prone to maintain organic debris—such as fallen leaves lining on soil—unremoved within the crop rows (Fig. 11). This differs from production fields, where the agricultural companies (e.g., Umoe BioEnergy) usually keep the crop area free of this material, as well as the plants tend to be less dry due to more frequent irrigation/rain. These distinct practices also have a pronounced effect on the quality of the IR images collected from the experimental crop rows (Fig. 12), in which the cluttered sets of terrain and leaves result in decreased thermal contrast between them.

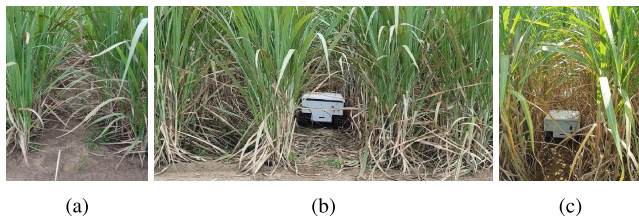


FIGURE 11. Unremoved organic debris observed in different sugarcane crop row tunnels at Embrapa Agrobiology experimental field - (a) crop row without the robot; (b) robot near the row entrance; (c) robot further inside the row, with less sunlight and more debris present on the ground.

A set of odometry-based, Cartesian regulation control approaches in closed-loop is also implemented on the robotic system and tested in the experimental field, with

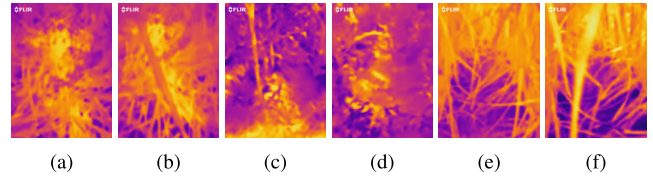


FIGURE 12. IR images of sugarcane crops in the experimental field at Embrapa Agrobiology, where the thermal contrast is less evident than in the production field. Images (a-d) for short canes, with soil warmer than plants. Images (e-f) for taller canes, with soil colder than plants.

the aim of preliminary investigating how the odometry error accumulation can interfere with the robot motion considering its structure and the sand/clay operational terrain. Here, we consider the problem of designing a feedback controller to drive TIBA robot from its current position $p = [x \ y]^T$, with respect to a world frame, to a given Cartesian position $p_d = [x_d \ y_d]^T$ without specifying the final orientation ϕ , as per [51]. In this context, the regulation control law defined by the input vector $\eta = [v_x \ \omega]^T$, where:

$$\begin{aligned} v_x &= k_1 (e_x \cos(\phi) + e_y \sin(\phi)), \\ \omega &= k_2 (\text{atan2}(e_y, e_x) - \phi), \end{aligned} \quad (3)$$

can be used as robot inputs to guarantee that the Cartesian errors, $e_x := x - x_d$ and $e_y := y - y_d$, converge to zero asymptotically by selecting $k_1, k_2 > 0$. For a stability analysis, please consult [51]. It is important to remark that k_1 acts directly on the robot's linear velocity, while k_2 forces the actuation on its angular velocity against the system's nonholonomic constraint. This effect can be mitigated if k_2 is selected to be smaller than k_1 .

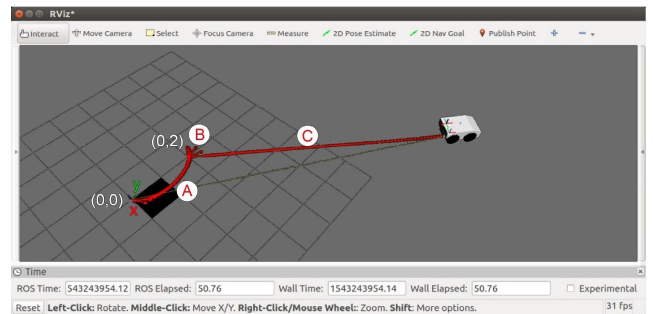


FIGURE 13. Embrapa Agrobiology experimental field, Seropédica-RJ, Brazil - Motion control on uneven terrain of crop row tunnels. RViz environment showing a practical example of the robot moving Phases A, B and C.

This approach was tested for two situations: (i) robot chassis rested in support, with wheels rotating but not touching terrain; (ii) robot operating on the uneven terrain of the crop row tunnels in the Embrapa Agrobiology experimental field. In both cases, odometry results were gathered from the generated ROS .bag files and visualized in the RViz environment. For the first case (free wheels), average position error accumulation in x - and y -axes was estimated in 1 cm per 7 m traveled. In contrast, for the second case (robot on terrain), this error accumulation was estimated in 1 cm per 4 m traveled. Finally, an animation in

the RViz environment was set for a practical example in which TIBA robot follows a certain trajectory in the sugarcane field composed of three phases. Fig. 10a-c show the robot in the field during Phases A, B, and C, respectively, while Fig. 13 displays the same motion as visualized in RViz. They are defined as: (i) **Phase A**, position regulation in closed-loop, driving the robot from initial position $p_0 = [0 \ 0]^T$ to target position $p_d = [0 \ 2]^T$, using the regulation kinematic control scheme described in Eq. (3), and control gains $k_1 = 0.5, k_2 = 0.3$; (ii) **Phase B**, proportional closed-loop orientation control, steering the robot from initial orientation $\phi_0 = 0$ to target orientation $\phi_d = \pi/4$ with in-place rotation, by selecting $v_x = 0$ and $\omega = -k_2 e_\phi$, where the chosen gain $k_2 = 0.3 > 0$ guarantees asymptotic convergence of the orientation error $e_\phi := (\phi - \phi_d)$ to zero; (iii) **Phase C**, remote control following a straight path along the crop row tunnel, with $v_x = 0.35$ and $\omega = 0$. All units are expressed in SI (m, rad, m s^{-1} , and rad s^{-1}).

C. CROP ROW FOLLOWING USING THERMAL IMAGING

As discussed in the authors' previous work [38], the quality of the thermal images collected in the field (Subsection II-B3) motivates the development of a crop row following approach that utilizes thermal IR images as input for a localization algorithm that locally estimates a reliable and accurate reference (i.e., the row centerline) that the robot shall follow. Given the requirement of slow speeds along the sugarcane crops (i.e., $\approx 1.26 \text{ m s}^{-1}$), there is enough room for testing image processing via deep learning to support identifying the row centerline. As discussed in Section I, thermal IR imaging combined with deep learning has already been investigated for several agricultural applications, but not for the crop row following problem presented in this work. Here, we present an IR image-based crop row following scheme by unveiling its main concepts and different implementation modules (Fig. 14).

In the localization and path planning module, our goal is to obtain the robot localization with respect to the row centerline. Since we seek a simple and inexpensive strategy to make the robot follow a straight line, it is unnecessary to find its position on the x -axis along the corridor path. This allows us to split the localization module in three steps or phases:

- (i) *Odometry update*: Based on feedback from odometry data and robot's kinematic model, it is one of the methods used for localization, along with IR imaging. Its connection to the control scheme is shown in Subsection III-A1.
- (ii) *Centerline feature extraction*: Use of thermal imaging to identify—via an image segmentation technique—the triangular-shaped feature that represents the path ahead, similar to a runway in perspective, and the row centerline that leads to the path vanishing point.
- (iii) *Centerline offset estimation*: Estimation and quantification of the robot's offset displacement e_y and misalignment e_ϕ relative to the row centerline, by using

the output from the previous phase and robot/camera parameters.

The robot localization and path planning phase is critical to allow the robot to operate under various conditions, such as image noise, partial camera lens occluded by leaves, uneven terrains causing the robot rotation around the x and y -axis, and image processing errors. This module, which is based on AI model and IR imaging, was briefly introduced in [38]. Here, we will provide a recap of the CNN-based centerline feature extraction process (in Subsection II-C2), and a more detailed outline of the odometry update and centerline offset estimation steps (in Subsections II-C1 and II-C3, respectively).

It is worth highlighting that the central contribution of this work is the introduction of a new module—a robust sliding mode control scheme—which will be addressed in Section III. This module will be validated through simulations and comparisons with other control methods within a novel computer environment developed specifically for this purpose using data obtained from real field conditions. Note that this work does not address the problem of crop rows switching using autonomous maneuvers in headlands (i.e., turnrow areas), which will be investigated in future research.

1) ODOMETRY UPDATE

Odometry is a core component of robot localization, providing the system with estimates of its pose based on the integration of motion data. In this context, it uses feedback from the robot's kinematic model, allowing the system to update its pose as it navigates through the environment. The odometry module integrates wheel encoder data (or simulated velocity information in the case of numerical simulations) to estimate changes in the robot's pose $q(t)$ over time. While odometry alone will accumulate errors due to wheel slippage or uneven terrain, it serves as a complementary method for localization, especially when fused with external sensors (such as IR cameras, the focus of this work), as to be detailed in the control scheme, Subsection III-A1. For numerical simulations, the odometry feedback is not acquired from wheel encoders, but obtained from integration of the kinematic model and the robot velocities. In this work, to obtain an exact reconstruction of q_{k+1} under the assumption of constant velocity inputs within the sampling interval, we introduce the following equation for the odometric localization update [51]:

$$\begin{cases} x'_{k+1} = x'_k + \frac{v_k}{\omega_k} \Delta_{\sin} + x_{icr} \Delta_{\cos}, \\ y'_{k+1} = y'_k - \frac{v_k}{\omega_k} \Delta_{\cos} + x_{icr} \Delta_{\sin}, \\ \phi'_{k+1} = \phi'_k + \omega_k h, \end{cases} \quad (4)$$

where $h = t_{k+1} - t_k$ is the sampling interval, $\Delta_{\sin} = \sin(\phi'_{k+1}) - \sin(\phi'_k)$, and $\Delta_{\cos} = \cos(\phi'_{k+1}) - \cos(\phi'_k)$.

2) CENTERLINE FEATURE EXTRACTION

As introduced in [38], in this phase, the first goal is to explicitly identify the “hot” areas of the thermal image,

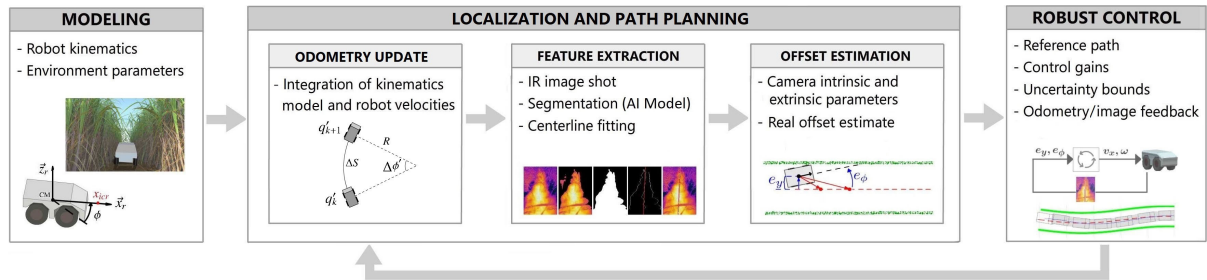


FIGURE 14. Framework for the thermal IR imaging based crop row following strategy - Modeling, localization, path planning, and robust control.

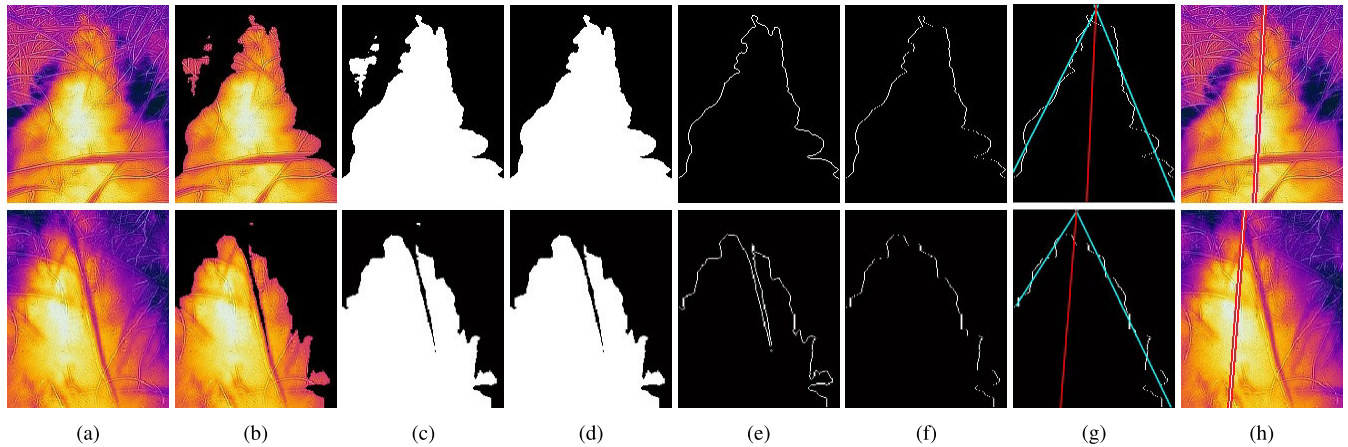


FIGURE 15. Steps of centerline fitting after color thresholding (CT) applied - (a) original image; (b) “runway” extracted via CT; (c) binary conversion of the extracted feature; (d) smaller pixel groups removed; (e) contour extraction; (f) outer pixels per row; (g) linear regression on left and right sides separately, and mean line at center; (h) final result.

which are believed to match with the soil as observed from the field experiment. From the temperature color scale utilized in the collected images from the field, the hot temperatures lie within a color spectrum around yellow, orange, red, and white. Notice that, in Fig. 9, the IR grayscale image can also be a valid choice to be considered in future works to simplify image processing due to the direct correspondence between temperature and gray level. By empirically analyzing all the collected media, we could observe that setting a basic color threshold (CT)—set to a hue range within $\mathcal{H} \in [0.145, 0.944]$, by using the Hue-Saturation-Value (HSV) color representation—would be sufficient to extract the ‘runway’ feature for most of the pictures.

In Fig. 15a to 15f, we can observe from two real field image examples how the extracted ‘runway’ feature—is expected to resemble a triangular or trapezoidal polygon, can be used as an input to estimate a straight line that represents the “center” of this feature. In [38], this was achieved via linear regression of both sides of the lane feature contour, and by taking the mean pixel of both contours at every row, as shown in Fig. 15g and 15h. Results from the feature extraction application for several examples can be found in Fig. 16.

We may not extract a clear and reliable ‘runway’ feature only by using a hot-ground/cold-plant color threshold, especially for scenarios with: (i) many leaves occluding the soil; (ii) colder environments, when daylight is reduced,

or where sunlight reaching soil is attenuated by taller cane stalks, creating the situation where the soil is colder, and its temperature difference to the cane stalks is lower, nonexistent, or even opposite.

In this context, as mentioned in Section I, deep convolutional neural networks (DCNNs) or deep CNNs, arise as a potential solution to overcome the aforementioned situations with expected low computation costs due to the slow navigation speed adopted in the sugarcane crop. In our case, we want to train a CNN model to delimit a polygon over the “hot” runway, even if this feature has an unusual shape or is blurry and overshadowed by a small temperature difference from the background, noisy, or “cold” (Fig. 17). To conduct a preliminary verification of the proposed approach and utilize a sufficient amount of collected field images, A CNN-based model (Fig. 18) was trained with IR thermal images using a 10-layer SegNet VGG16, with the following architecture: 1-Image Input (image size $96 \times 72 \times 3$ pixels), 2-Conv. 2D (3 channels, 400 filters, size 3×3 , stride 1×1 , unit padding size 4D), 3-ReLU, 4-Max. Pooling 2D (pool size and stride 2×2 , zero-padding), 5-Conv. 2D (400 channels, 400 filters, size 3×3 , stride 1×1 , unit padding size 4D), 6- ReLU, 7-Transp. Conv. 2D (400 channels, 400 filters, size 4×4 , stride 2×2 , unit cropping size 4D), 8-Conv. 2D (400 channels, 2 filters, size 1×1 , stride 1×1 , zero-padding), 9-Softmax, and 10-Output Pixel Classification (2 categories, image size $96 \times 72 \times 3$ pixels).

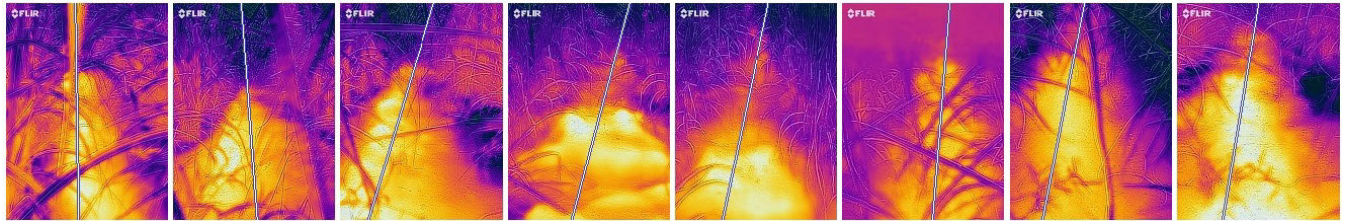


FIGURE 16. Examples of centerline fitting after applying color thresholding (CT). The criteria utilized to determine the success of the centerline fitting were partially subjective. However, all pictures were checked to see whether the line crossed or not the “runway” peak and whether it was kept within the “runway” area. The failure cases will be omitted here for the space-saving purpose, but these can be found in [38].

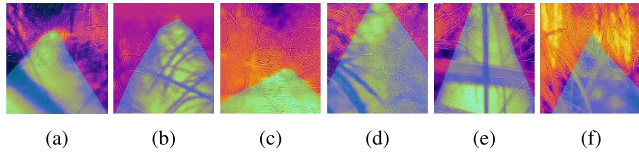


FIGURE 17. Examples of bad situations and expected features - (a) erratic curved shape; (b) blurry; (c) small color difference; (d) noisy vision/many branches; (e) leaves trapping occluding image; (f) colder environment or tall plants.

An initial set of 180 downsampled and annotated IR images was carefully selected from the dataset to train the network. Horizontal flipping and shifting were applied to this set to augment the data, resulting in 540 training samples. Moreover, 360 independent IR images were selected exclusively for evaluation, with 180 assigned for validation during training and 180 for final testing and performance metrics assessment. This resulted in a dataset of 900 IR images—captured in both the production and experimental crops—with diverse environmental, crop and geometry conditions (as summarized in Table 1). The IR thermal image dataset was split into 60% for training, 20% for validation, and 20% for testing. Correspondingly, the expected ground truth outputs were defined as binary images, representing two distinct categories: ‘background’, denoted by black or 0, and ‘runway’, denoted by white or 1. During this initial investigation, various CNN hyperparameters (kernel widths, strides, pooling, and padding values) were systematically assessed. Future work will encompass a more sophisticated exploration of this segmentation problem, aiming for strategies that optimally leverage the ability of CNN models to cope with the situations found inside the crop rows.

As expected, the obtained result from this preliminary training yielded potential contributions in the context of the ‘runway’ feature detection. Fig. 19 shows the outcomes for some examples from the dataset that were submitted to the trained CNN model. Note that, specifically for the two highlighted instances in Example 1 and 2 (see Fig. 19a), the leaf occlusion issue was satisfactorily circumvented for CNN as shown in Fig. 19c. In contrast, the color thresholding (CT) algorithm resulted in the fragmentation of the image feature into smaller regions (Fig. 19b). After training the CNN model for up to 100 epochs with a learning rate of 10^{-3} , its performance was evaluated on the test set by comparing pixel-wise the predicted segmentation results

TABLE 1. Dataset diversity across the two sugarcane field sites for IR thermal image collection - Umoe BioEnergy Production Field and Embrapa Agrobiology Experimental Field. The table shows key environmental and geometry characteristics observed at each site during data collection.

Description	Umoe Field	Embrapa Field
Crop type	Production	Experimental
Season	Summer	Spring
Data collection time	3–7 PM	10 AM–4 PM
Terrain	Uneven	Flatter
Ground condition	Clean	Covered with leaves
Soil type	Oxisol	Planosol
Plant height	1–3 m	2 m
Crop row width	1.5 m (avg.)	0.75 m (avg.)
Row layout	Curved	Straight
Row length	>500 m	60–80 m

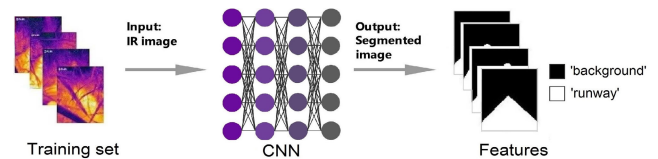


FIGURE 18. CNN model trained to segment soil “runway” from IR thermal images, based on a 10-layer SegNet VGG16: 1-Image Input (image size $96 \times 72 \times 3$ pixels), 2-Conv. 2D (3 channels, 400 filters, size 3×3 , stride 1×1 , unit padding size 4D), 3-ReLU, 4-Max. Pooling 2D (pool size and stride 2×2 , zero-padding), 5-Conv. 2D (400 channels, 400 filters, size 3×3 , stride 1×1 , unit padding size 4D), 6-ReLU, 7-Transp. Conv. 2D (400 channels, 400 filters, size 4×4 , stride 2×2 , unit cropping size 4D), 8-Conv. 2D (400 channels, 2 filters, size 1×1 , stride 1×1 , zero-padding), 9-Softmax, and 10-Output Pixel Classification (2 categories, image size $96 \times 72 \times 3$ pixels).

with the ground truth, resulting in the confusion matrix in Table 2a. In this evaluation, the ‘runway’ category (region of interest) was treated as the “positive” classification, and the ‘background’ category as the “negative” classification. Consequently we could calculate relevant standard metrics for the model performance: accuracy of 84.3%, precision of 86.5%, recall of 82.1%, and F1-score of 0.84 (see Table 3). This result is considered satisfactory for this initial analysis with a limited dataset in a noisy agricultural environment, while the resulting feature uncertainties will be modeled as perturbations in the robust control design, as detailed in the control methodology section. To preliminarily estimate the computational cost associated with the CNN model compared

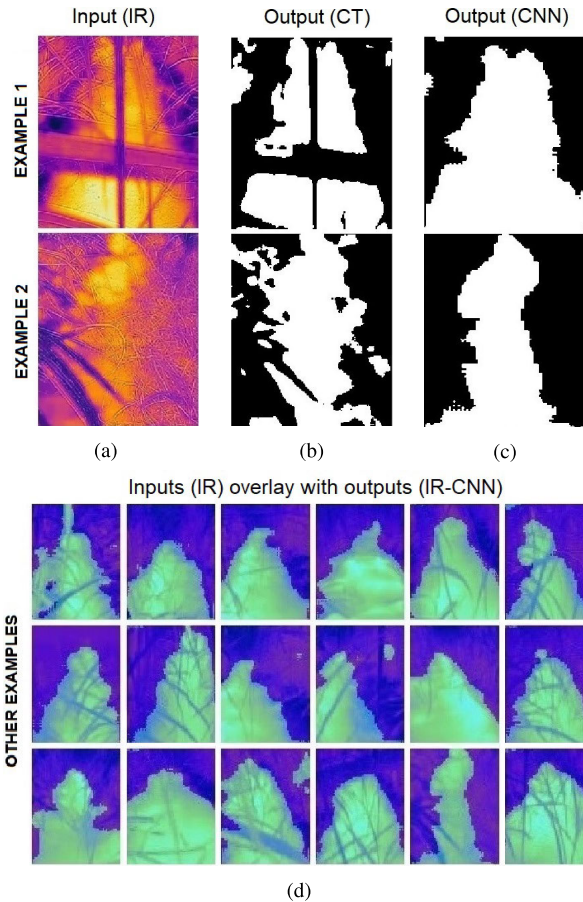


FIGURE 19. Examples from the trained CNN model - (a) IR thermal image input; (b) “runway” feature output by CT; (c) “runway” feature output by CNN; (d) collection of results from CNN displayed by overlaying output over more input examples.

to a simple image processing algorithm such as CT, both methods were applied to the same evaluation set. The mean and standard deviation of the elapsed processing time were calculated as $\mu_{ct} = 0.37$ ms and $\sigma_{ct} = 0.16$ ms for the CT algorithm, and $\mu_{cnn} = 815.90$ ms and $\sigma_{cnn} = 175.80$ ms for the CNN model. Note that the mean for CNN is about 2,000 times higher than the average obtained for CT but still satisfactorily lower for the required robot velocity and expected detection rate for the crop row following task. This assessment was conducted using a computer equipped with Intel[®] Core[™] i7-7700HQ CPU @ 2.80 GHz running Linux Ubuntu 20.04, 16GB RAM and nVIDIA GeForce[™] GTX-1050 4GB GDDR5 GPU. We can conclude that the fusion of the CNN model outcomes with linear regression noticeably enhances the estimation of the crop row centerline, as demonstrated in the two examples in Fig. 20.

The employment of a CNN model robustly accentuates the superior performance of using thermal IR images in comparison to conventional RGB images. Notably, as discussed in this work, the thermal radiation phenomenon inherent in the IR images collected in sugarcane crop rows offers features (“runway”) that are markedly clearer and more discernible for image processing than those derived from

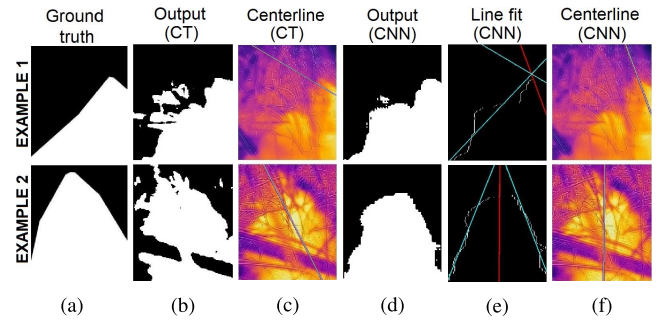


FIGURE 20. Result from centerline fitting after using the CNN model - (a) “ground truth” for expected “runway” feature; (b) output “runway” after CT applied; (c) centerline after CT applied; (d) output “runway” after CNN applied; (e) centerline fitted via linear regression after CNN applied; (f) centerline after CNN applied.

RGB cameras. This pronounced visibility arises from the distinct heat absorption and emission properties between soil and plants. Moreover, this clarity is advantageous as it tends to remain consistent upon varying incident daylight. This assessment is compellingly illustrated in Fig. 21, which showcases the compared outputs of two CNN models: one being trained only with IR thermal images (i.e., the same CNN architecture previously trained), and the other trained with the corresponding RGB images of the same locations, and harmonized with identical network parameters. This was made feasible owing to the design of the FLIR[®] One V2 camera, which incorporates both RGB and IR sensors. The confusion matrix corresponding to the CNN trained with RGB images is shown in Table 2b, from which the following performance metrics were obtained: accuracy of 56.7%, precision of 59.2%, recall of 92.8%, and F1-score of 0.72 (see Table 3). Compared to the model trained only with IR images, we can observe that the RGB-CNN underperformed in terms of accuracy and precision. This high recall value could be explained by the tendency of the RGB-CNN model to oversegment the true ‘runway’ pixel area. However, since ‘runway’ pixels represent only a small proportion of the image, this also results in a high number of false positives, as many ‘background’ pixels are misclassified. It is worth highlighting that the networks discussed so far were trained exclusively with “hot runway” images (hot ground, cold plants). Therefore, the segmentation of the “cold runway” images (cold ground, hot plants), while not the primary focus of this work, presents an opportunity for future exploration. A more intricate learning network could be developed to incorporate a broader and more diverse training dataset, as previously discussed in Subsection II-B3. Notably, the authors [38] already proposed and tested a learning approach in which a secondary CNN was introduced to classify images as hot or cold before segmenting them in a subsequent stage, using two separate CNNs trained on distinct ‘hot’ and ‘cold’ datasets.

As verified in the experiments, CNN models have demonstrated considerable appeal to handle IR image segmentation in the sugarcane environment, offering relevant potential for future improvements in inference accuracy and

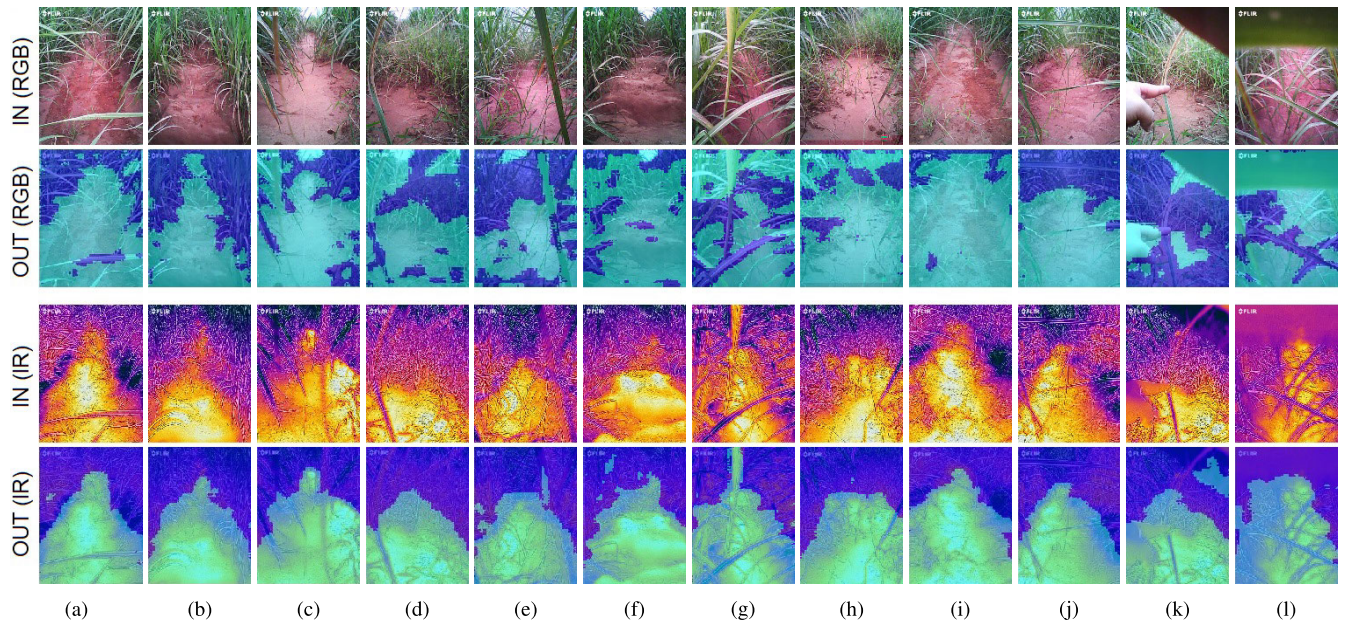


FIGURE 21. Image comparison of “runway” output after CNN application with RGB versus IR inputs, taken from the same camera at the same places. **Case analysis:** (a-b) CNN-RGB with acceptable output, but worse than CNN-IR; (c-h) CNN-RGB fails and provides noisy output; (i-j) CNN-RGB fails and provides one big feature group; (k-l) occluding leaf at image top and intruding human hand at left, both satisfactorily ignored by CNN-IR, but misled CNN-RGB.

computational cost reduction. In this preliminary evaluation, several details were overlooked but must be addressed in the next developments, such as: (i) use of IR images in grayscale to minimize processing cost and the number of CNN kernels; (ii) removal of RGB camera overlay from the IR image aiming to prevent undesired patterns misleading the training process for seeking textures rather than colors; (iii) use of landscape format to cover a broader field-of-view; (iv) early conversion of inputs to alternate color spaces (e.g., OHTA); (v) training of more newer and hybrid networks interlaced with other navigation phases—such as vehicle control and row switching—which would require the definition of alternative input type—e.g., videos—and feature classes, such as ‘robot pose’, ‘safe reference trajectory’, ‘alarm near collision’, ‘curved lane’, among others; (vi) integration of multiple cameras mounted on different points on the robot chassis. Arguably, one particularly promising strategy is to employ deep learning to detect crop rows as lanes (i.e., identifying the ‘kerbs’ as features), rather than focusing on the ‘runway’ shape segmentation, as evidenced by relevant existing work mentioned in Section I.

In the sugarcane crops case study, identifying corridor sidelines may be a smarter strategy than trying to segment a pixel region because of two main motivations. Firstly, it enables the use of both ‘hot’ and ‘cold’ ground images in the training set, as the CNN model would strive to identify a border with the indication of rapid color variation, in spite of identifying which color belongs to the ‘runway’ region or the plant region. Secondly, the knowledge of the corridor limits configures additional information for devising strategies for obstacle avoidance during the trajectory planning, which was not provided by a simple image segmentation of the ‘runway’.

TABLE 2. Confusion matrices for the trained CNNs. A total of 1,244,160 pixels were analyzed, corresponding to 180 test images of size 96×72 . The terms TP, FN, FP, and TN stand for true/false positives/negatives, “runway” for the “positive” category, and “background” for the “negative” category.

(a) CNN trained with IR images.

	Predicted runway	Predicted background
Actual runway	TP = 42.1%	FN = 9.1%
Actual background	FP = 6.6%	TN = 42.2%

(b) CNN trained with RGB images.

	Predicted runway	Predicted background
Actual runway	TP = 56.5%	FN = 4.4%
Actual background	FP = 38.9%	TN = 0.2%

TABLE 3. Performance metrics of the trained CNNs, calculated from confusion matrices using the following definitions:

$Acc = (TP + TN) / (TP + TN + FP + FN)$, $Prec = TP / (TP + FP)$, $Rec = TP / (TP + FN)$, and $F1 = 2PR / (P + R)$.

Metric	CNN (IR)	CNN (RGB)
Accuracy (Acc)	84.3%	56.7%
Precision (Prec)	86.5%	59.2%
Recall (Rec)	82.1%	92.8%
F1-score (F1)	0.84	0.72

3) CENTERLINE OFFSET ESTIMATION

The goal of this phase is to estimate the robot offset from the crop row centerline by verifying the following steps: (i) compute the position vector $p_{ir} \in \mathbb{R}^3$ —expressed in the robot frame $\mathcal{F}_r = \{\vec{x}_r, \vec{y}_r, \vec{z}_r\}$ —of a target object

represented by an arbitrary point P_i lying on the ground, given its projected image coordinates $M_{is} = (x_{is}, y_{is})$; (ii) define a straight line from two arbitrary different points, p_{1r} and p_{2r} , and by using a proper mathematical relationship, calculate the robot offset pair (e_y, e_ϕ) . In other words, we have to solve the following set of equations:

$$p_{ir} = f_1(M_{is}), \quad i = 1, 2, \quad (e_y, e_\phi) = f_2(p_{1r}, p_{2r}), \quad (5)$$

where $f_1(\cdot)$ and $f_2(\cdot)$ are nonlinear functions to be properly designed in what follows. Here, we consider that the robot offset pair is defined by the composition of the lateral displacement and misalignment angle between the sagittal axis of the robot and the corridor ground centerline, denoted by e_y and e_ϕ , respectively.

To describe this phase in detail, let us consider the image projection problem and the frames depicted in Fig. 22, where the coordinates of the target object P_i is denoted by the position vector $r_i = [x_i \ y_i \ z_i]^T$ expressed in the camera frame $\mathcal{F}_c = \{\vec{x}_c, \vec{y}_c, \vec{z}_c\}$. Here, without loss of generality, we assume that the centroid of the target object can be chosen as the arbitrary point P_i , and that the IR sensor can be modeled as a *pinhole* camera, which is widely used for thermal camera calibration, as per [38] and [56]. Notice that the position vector $r_i \in \mathbb{R}^3$ always crosses the inner limits of the *image plane* or *sensor plane*, which consists of a rectangular array of photosensitive elements, termed photosites or pixels, with dimensions of α_x (width) and α_y (height) measured in millimeters.

The rectangular array of photosites can be interpreted as a virtual plane, being the dimension of the image projected onto it limited by the camera lens aperture. The projected image of the arbitrary point P_i , denoted by the point M_i , can be expressed in the *projection frame* $\mathcal{F}_p = \{\vec{x}_p, \vec{y}_p, \vec{z}_p\}$ by the 3-dimensional vector $M_{ip} = [x_{ip} \ y_{ip} \ 0]^T$ or in the *sensor frame* $\mathcal{F}_s = \{\vec{x}_s, \vec{y}_s, \vec{z}_s\}$ by the 2-dimensional vector $M_{is} = [x_{is} \ y_{is}]^T$. It is worth mentioning that the normalized image-plane coordinates, x_{ip} and y_{ip} , can be calculated respectively as:

$$x_{ip} = f \frac{x_i}{z_i}, \quad y_{ip} = -f \frac{y_i}{z_i}, \quad (6)$$

where $f > 0$ is the focal length of the camera lens. Note that, pixels are uniformly sized and centered on a regular grid. Therefore, the normalized image-plane coordinates are related to the pixel coordinates, x_{is} and y_{is} , respectively as:

$$x_{is} = \frac{x_{ip}}{\rho_h} + x_{s0}, \quad y_{is} = \frac{y_{ip}}{\rho_w} + y_{s0}. \quad (7)$$

Here, (ρ_h, ρ_w) denote the pixel size, the height and width of each pixel respectively, and (x_{s0}, y_{s0}) denote the camera principal points, the coordinates where the principal planes intersect with the optical axis.

The origin of the sensor frame $\mathcal{F}_s$ is located at the top-left hand corner of the sensor plane, where the photosensitive sensor lies. Then, the top-left pixel is $M_{is} = (1, 1)$. The origin of the projection frame $\mathcal{F}_p$ is located at the image center. If we consider the sensor pixel resolution of β_x (columns or

“height”) versus β_y (rows or “width”) measured in pixels, the dimension of a single pixel is then defined by $\rho_h = \alpha_x/\beta_x$ and $\rho_w = \alpha_y/\beta_y$, and the camera principal points defined by $x_{s0} = \beta_x/2$, $y_{s0} = \beta_y/2$. Then, we can define the relation between M_{is} expressed in pixel coordinates and its corresponding M_{ip} represented in normalized image-plane coordinates as:

$$M_{ip}(x_{is}, y_{is}) = [\rho_h(x_{is} - \beta_x/2) \ \rho_w(y_{is} - \beta_y/2) \ 0]^T. \quad (8)$$

Notice that, in general, the camera frame $\mathcal{F}_c$ is aligned with the center of the sensor plane and $(\beta_x/2, \beta_y/2)$ are the coordinates of the point where the camera optical axis intersects the sensor plane, the camera principal point. The camera frame $\mathcal{F}_c$ is located in the coordinates $c = (c_x, c_y, c_z)$ expressed in the robot frame $\mathcal{F}_r$, the reference frame for suitably describing the robot’s motion and lying on the ground level. The misalignment angles between frames $\mathcal{F}_c$ and $\mathcal{F}_r$ are given by $\theta = (\theta_r, \theta_p, \theta_y)$, represented in RPY/Euler angles. The image projection problem consists of finding the coordinates of a set of arbitrary points P_i for $i = 1, 2, \dots, n$ from a given input image, generated in the previous step (see Section II-C2). Such an image provides the result of the linear regression method applied over the original thermal image, from which we extract a set of points M_{is} for $i = 1, 2, \dots, n$. It is worth mentioning that since the arbitrary points P_i are located at the target objects on the corridor ground centerline and expressed with respect to the robot frame $\mathcal{F}_r$, these are denoted by p_{ir} for $i = 1, 2, \dots, n$. Once the point M_{ip} is obtained, it is necessary to compute the object image projection M_{ir} referred to the robot frame $\mathcal{F}_r$, which is given by:

$$\tilde{M}_{ir} = T_{rc} T_{cp} \tilde{M}_{ip}, \quad (9)$$

being $\tilde{M}_{ip} = [M_{ip}^T \ 1]^T \in \mathbb{R}^4$ and $\tilde{M}_{ir} = [M_{ir}^T \ 1]^T \in \mathbb{R}^4$. Notice that, T_{ij} is the homogeneous transformation matrix denoting the pose of the frame $\mathcal{F}_j$ with respect to the frame $\mathcal{F}_i$ for $i = \{r, c\}$ and $j = \{c, p\}$, respectively. Thus, the coordinate transformations $T_{rc} = T_{rc'} T_{c'c}$ and T_{cp} yield, respectively:

$$T_{rc} = \begin{bmatrix} R_y(\pi/2) p_{rc} & 1 \\ 0_{1 \times 3} & 1 \end{bmatrix} \begin{bmatrix} R_{c'c} & 0_{3 \times 1} \\ 0_{1 \times 3} & 1 \end{bmatrix}, \quad (10)$$

$$T_{cp} = \begin{bmatrix} R_x(\pi) [0 \ 0 \ f]^T & \\ 0_{1 \times 3} & 1 \end{bmatrix}, \quad (11)$$

where $p_{rc} = [c_x \ c_y \ c_z]^T$ is the position vector of the frame $\mathcal{F}_c$ with respect to the frame $\mathcal{F}_r$, $R_x, R_y \in \mathbb{SO}(3)$ are elementary rotation matrices around the x - and y -axis, respectively, and $R_{c'c} \in \mathbb{SO}(3)$ is a composite rotation matrix which comprises the case where the camera frame $\mathcal{F}_c$ is not aligned with the robot frame $\mathcal{F}_r$. For simplification purposes, in this work, we consider the simplest case in which $R_{c'c} = I_3$.

Now, to compute the location of the point P_i with respect to the robot frame $\mathcal{F}_r$, denoted by the position vector p_{ir} , let us first define the position vector ℓ_i that connects the camera frame origin $\mathcal{O}_c$ to the image projection point M_i (Fig. 23a) and it is expressed in the robot frame $\mathcal{F}_r$ as $\ell_{ir} = M_{ir} - p_{rc}$.

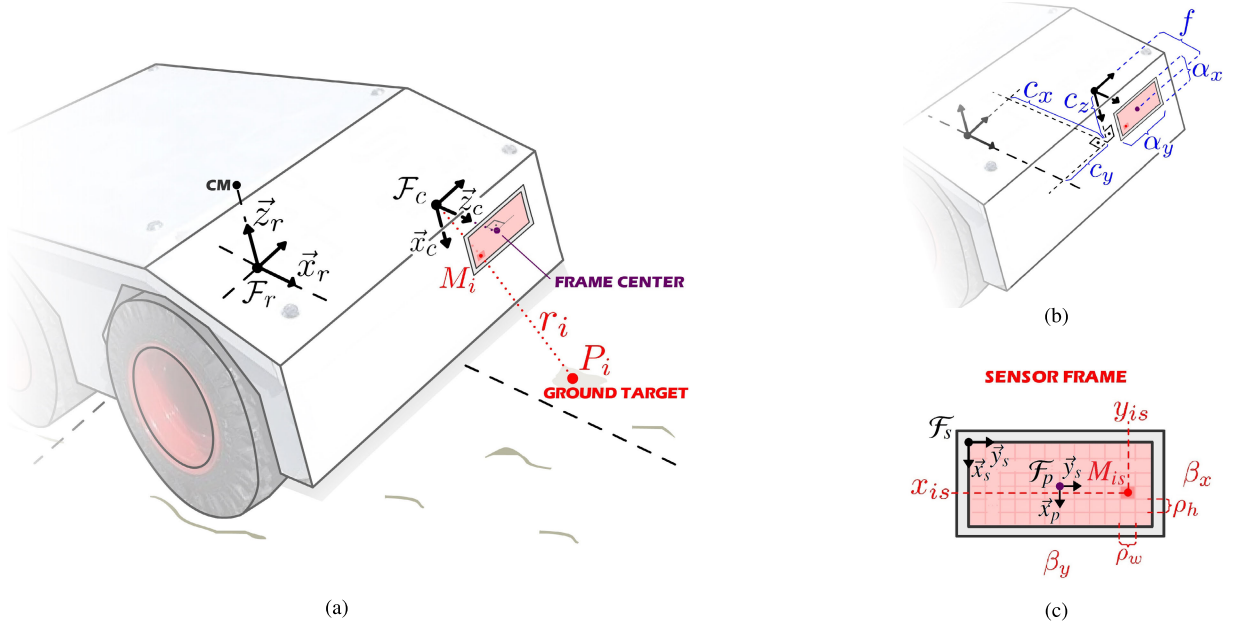


FIGURE 22. Camera model and relevant frames - (a) robot and coordinate frames in perspective, $\mathcal{F}_r$ (robot frame), $\mathcal{F}_c$ (camera frame), $\mathcal{F}_p$ (projection frame); (b) camera parameters; (c) $\mathcal{F}_s$ (sensor frame) and image coordinates.

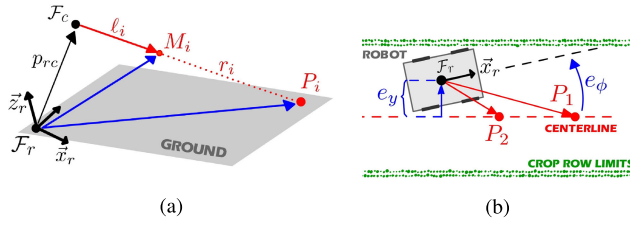


FIGURE 23. Centerline offset estimation geometry - (a) position of a target object P_i with respect to the robot frame $\mathcal{F}_r$; (b) robot offset (e_y, e_ϕ) from the crop row centerline.

If we extend the position vector ℓ_i directly along the line r_i by using a multiplier factor $\gamma_i \in \mathbb{R}$, we obtain the following equation:

$$p_{ir} = p_{rc} + \gamma_i \ell_{ir}. \quad (12)$$

Considering both the robot base and the ground aligned, we observe that the z -coordinate of vector p_{ir} will be always zero, since point P lies on the ground and limits the elongation of $\gamma_i \ell_{ir}$. Thus, we can re-write Eq. (12) only looking into the vectors z -coordinates (using operation $\hat{k}^T$ to extract these⁴) and replacing $p_{rc} = M_{ir} - \ell_{ir}$, which yields:

$$0 = \hat{k}^T p_{ir} = \hat{k}^T (M_{ir} - \ell_{ir}) + \gamma_i \hat{k}^T \ell_{ir} \quad (13)$$

with $\gamma_i = -(\hat{k}^T M_{ir})/(\hat{k}^T \ell_{ir}) + 1$. Thereby, it is finally possible to find the robot offset $e = [e_y \ e_\phi]^T$ from the corridor centerline (Fig. 23b) with respect to an inertial frame $\mathcal{F}_0$ whose x_0 -axis is aligned with the sagittal axis of the

⁴Here, it is worth defining the elementary unit vectors as $\hat{i} = (\bar{x}_\psi)_\psi = [1 \ 0 \ 0]^T$, $\hat{j} = (\bar{y}_\psi)_\psi = [0 \ 1 \ 0]^T$ and $\hat{k} = (\bar{z}_\psi)_\psi = [0 \ 0 \ 1]^T$, where ψ represents the index of any arbitrary reference frame $\mathcal{F}_\psi$.

mobile robot. In this work, we consider the computation of two arbitrary points p_{2r} and p_{1r} previously obtained from M_{1s} and M_{2s} , respectively, wherein both points obligatory belonging to the image runway triangle border and the straight line fitted by the linear regression phase. We then compute the vector:

$$v_p = p_{1r} - p_{2r}, \quad (14)$$

which lies along the straight line defined by both points. Notice that, from Fig. 23b, we assign point p_{1r} to be ahead of point p_{2r} , so that the vector v_p will also point towards the direction of the robot motion. The lateral displacement e_y that defines the minimal distance from the origin of the frame $\mathcal{F}_r$ to the corridor centerline is the norm of the vector:

$$v_\perp = (v_p v_p^T) / (\|v_p\|^2) p_{1r} - p_{1r}, \quad (15)$$

which is orthogonal to the projection of the point p_{1r} onto the corridor centerline, that is:

$$e_y = \text{sgn}(\hat{j}^T v_\perp) \|v_\perp\|, \quad (16)$$

and has a positive sign if it is pointing to the positive direction of $\bar{y}_0$. Then, the angular misalignment e_ϕ is given by:

$$e_\phi = \text{sgn}(\hat{k}^T \omega_p) \cos^{-1}(\hat{i}^T v_p / \|v_p\|), \quad (17)$$

where $\text{sgn}(\cdot)$ denotes the sign function and $\omega_p = v_p \times \hat{i}$ is a cross product operation. Finally, after combining the previous equations, we can explicitly define the function $f_1(\cdot)$ in terms of the points $M_{is} = (x_{is}, y_{is})$ for $i = 1, 2$ as:

$$p_{ir} = f_1(x_{is}, y_{is}) = -\frac{1}{\rho_h \beta_x} \begin{bmatrix} c_z f \\ c_z \rho_w \bar{\beta}_y \\ 0 \end{bmatrix} + \begin{bmatrix} c_x \\ c_y \\ 0 \end{bmatrix}, \quad (18)$$

where $\bar{\beta}_x = (\beta_x/2) - x_{is}$ and $\bar{\beta}_y = (\beta_y/2) - y_{is}$, as well as the function $f_2(\cdot)$ in terms of the points p_{1r} and p_{2r} , both introduced in Eq. (5), as follows:

$$\begin{bmatrix} e_y \\ e_\phi \end{bmatrix} = f_2(p_{1r}, p_{2r}) = \begin{bmatrix} \text{sgn}(\hat{j}^\top u_p - p_{1r}) \|u_p - p_{1r}\| \\ \text{sgn}(\hat{k}^\top (v_p \times \hat{i})) \cos^{-1}\left(\frac{\hat{i}^\top v_p}{\|v_p\|}\right) \end{bmatrix}, \quad (19)$$

where $u_p = (v_p v_p^\top p_{1r}) / \|v_p\|^2$ with $v_p = p_{1r} - p_{2r}$.

III. ROBUST CONTROL SCHEME

Here, we consider the control design problem for the skid-steering mobile robot (SSMR), TIBA robot, to ensure the successful tracking of a reference path inside the sugarcane crop tunnel under external disturbances and parametric uncertainties. To achieve this, we first revisit the fundamental kinematic equations of the robotic system, already introduced in Subsection II-A. Hereafter, for simplification purposes, it is worth defining the following short notation: $s_\psi = \sin(\psi)$ and $c_\psi = \cos(\psi)$ for any arbitrary angle ψ . The matrix S is defined such that the relation $\dot{q} = S(q)\eta$ yields:

$$\begin{cases} \dot{x} = c_\phi v_x + x_{icr} s_\phi \omega, \\ \dot{y} = s_\phi v_x - x_{icr} c_\phi \omega, \\ \dot{\phi} = \omega, \end{cases} \quad (20)$$

where $q = [x \ y \ \phi]^\top$ is the robot pose, $\eta = [v_x \ \omega]^\top$ represents the robot's controllable inputs (linear and angular speeds, respectively), and $x_{icr} \in \mathbb{R}$ symbolizes the robot's instantaneous center of rotation (ICR). The main control objective is to ensure that TIBA robot tracks a reference path at constant speed along the tunnel, with desired pose defined as $q_d = [x_d \ y_d \ \phi_d]^\top$ and reference input defined as $\eta_d = [v_{dx} \ \omega_d]^\top$, for a given choice of v_{dx} and ω_d . To find a feasible solution for the tracking problem, the desired pose q_d needs to be admissible for the kinematic model described by Eq. (20), with the tracking error defined as $\tilde{q} = q - q_d = [\tilde{x} \ \tilde{y} \ \tilde{\phi}]^\top$. By considering Eq. (20) and taking the time derivative of $\tilde{q}$, we obtain the following kinematic error align:

$$\begin{aligned} \dot{\tilde{q}} &= S(q)\eta - S(q_d)\eta_d \\ &= g_1(q)v_x + g_2(q)\omega + g_0(q_d), \end{aligned} \quad (21)$$

where $g_1(q) = [c_\phi \ s_\phi \ 0]^\top$ and $g_2(q) = [x_{icr}s_\phi \ -x_{icr}c_\phi \ 1]^\top$ are vector fields and $g_0(q_d) = -S(q_d)\eta_d$ is the drift of the system caused by the reference input signals.

A. TRANSFORMATION TO CHAINED FORM

The key idea of transforming the kinematic model of a mobile robot, such as the SSMR described by Eq. (20), into a canonical form has been widely explored in the literature to solve planning and control problems using an efficient, systematic procedure [50], [55], as discussed in Section I. Leveraging the chained form transformation [47], [51], the mobile robot kinematics can be transformed to a modified optimal chained form [42] using the coordinates and control

variables transformation matrices, T_{zq} and $T_{u\eta}$, providing a smooth map $(\tilde{x}, \tilde{y}, \tilde{\phi}, v_{dx}, \omega) \mapsto (z_1, z_2, z_3, u_1, u_2)$. The matrix T_{zq} is defined such that the relation $z = T_{zq}(q, \tilde{q})\tilde{q}$ yields the following expression:

$$\begin{cases} z_1 = \tilde{\phi}, \\ z_2 = c_\phi \tilde{x} + s_\phi \tilde{y}, \\ z_3 = (-\tilde{\phi}c_\phi + 2s_\phi)\tilde{x} - (\tilde{\phi}s_\phi + 2c_\phi)\tilde{y} - 2x_{icr}\tilde{\phi}, \end{cases} \quad (22)$$

where the variables z_1 and z_2 are assigned according to an intuitive geometric interpretation, as shown in [42], in terms of the offset of the actual controlled robot—with pose q and velocities (v_x, ω) —with respect to the reference robot—with pose q_d and desired velocities (v_{dx}, ω_d) . While z_1 is given by the orientation offset, z_2 is interpreted as the projection length of the robot reference offset onto the attitude direction of the controlled robot. Now, the new control variable $u = (u_1, u_2)$ can be related to the control and reference input signals, η and η_d , by the control variables transformation matrix $T_{u\eta}$, which is given such that $u = T_{u\eta}(q, \tilde{q})\eta + G_0(\tilde{q}, \eta_d)$ provides:

$$\begin{cases} u_1 = \omega - \omega_d, \\ u_2 = v_x - (s_\phi \tilde{x} - c_\phi \tilde{y})\omega - (c_\phi v_{dx} - x_{icr}s_\phi \omega_d). \end{cases} \quad (23)$$

The coordinates transformation in Eq. (22) and the change in control variables in Eq. (23), when combined, results in the following 3-dimensional chained form system $\dot{z} = \tilde{S}(z)u$ [42], described in the expanded form as:

$$\begin{cases} \dot{z}_1 = u_1, \\ \dot{z}_2 = u_2, \\ \dot{z}_3 = u_1 z_2 - z_1 u_2 + \underbrace{2[(x_{icr} + z_2 - x_{icr}c_{z_1})\omega_d - s_{z_1}v_{dx}]}_{\zeta}, \end{cases} \quad (24)$$

where $u = [u_1 \ u_2]^\top$ is the controlled input signals (to be designed) and ζ denotes the system's drift behavior caused by the reference input signals. As demonstrated by [42] this drift term can be utilized as a whole parameter to design a generic control law, employing a time-varying kinematic oscillator together with a nonholonomic integrator. Although this control approach is said to stand out for its applicability to any trajectory, it may introduce complexity due to its time dependence and requirement for prior settings of at least five positive constant design parameters. More sophisticated robust control schemes based on the sliding mode paradigm have the potential to treat the drift term ζ as a unique disturbance with known boundaries. Finally, we can define the reverse transformation $T_{\eta u}$, based on [42], that maps the robot controlled input η from the new control input u —i.e., $\eta = T_{u\eta}^{-1}(q, \tilde{q})[u - G_0(\tilde{q}, \eta_d)]$ —as follows:

$$\begin{cases} v_x = u_2 + (s_\phi \tilde{x} - c_\phi \tilde{y})(u_1 + \omega_d) + c_\phi v_{dx} - x_{icr}s_\phi \omega_d, \\ \omega = u_1 + \omega_d. \end{cases} \quad (25)$$

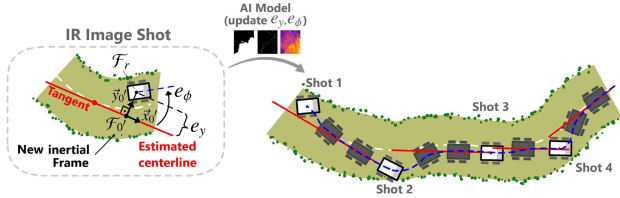


FIGURE 24. PBVS control scheme - Offsets are updated after the IR image shot (with period h_c) and image processing via CNN model. Red lines represent a “virtual straight path” to be followed after every IR image shot to update odometry. Blue line represents the mobile robot’s actual path.

1) POSE-BASED VISUAL SERVOING (PBVS) APPROACH WITH CAMERA-INDUCED DISTURBANCES

In the real-world scenario, as discussed in Section II-B2, the shapes of crop rows commonly found in sugarcane fields are not previously known and not perfectly straight, which requires the mobile robot to cope not only with uncertainty in modeling and terrain but also with curves in the corridor. These factors demand that the system constantly or periodically acquire and determine the robot’s pose with respect to the row centerline, as it constantly changes with respect to the world frame. In this paper, the continuous pose estimation conceptually aligns with the principles of a Pose-Based Visual Servoing (PBVS) system [54]. In this context, we reformulate the control problem considering a PBVS approach that combines the presented sliding mode control scheme with the IR thermal imaging system. Here, as per illustrated by Fig. 24, we consider that the world frame $\mathcal{F}_0$ is updated periodically—with period h_c —at every IR image shot by a new frame $\mathcal{F}_0'$, assigned⁵ so that the robot’s new pose is described by $q' = [x' \ y' \ \phi']^T$. Then, the robot restarts with the initial condition at $q'_0 = [x' \ e_y \ e_\phi]^T$, where (e_y, e_ϕ) are the robot offset estimates from the row centerline computed from the image processing algorithm, as shown in Fig. 23b, with the CNN model presented in Section II-C2.

For a generic corridor shape, as illustrated in Fig. 24, e_y is considered to be the minimum (orthogonal) distance of the robot to the prolongation of the virtual estimated row centerline, which is expected to be tangent to the curve located at the exact center of the same corridor. Now, we can redefine the tracking error as $\tilde{q}' = q' - q'_d$, where the reference pose remains defined as $q'_d(t) = q_d(t) = [v_{dx} \ t \ 0 \ 0]^T$, and the block diagram for a robust control scheme that considers IR thermal imaging, as described by Fig. 25.

Because IR thermal cameras often have inherently slow frame/refresh rates, when IR images are not available, it is required to close the control loop either with the odometry feedback or via sensor fusion (e.g., Kalman Filter approach)

⁵Note that the conceptual idea of assigning and updating the world fixed frame $\mathcal{F}'_0$ is similar to the kinematic modeling using a Frénet frame [50], which defines a generic point P on the mobile robot and a frame $\mathcal{F}_f$ over a curve C (the reference trajectory to be followed) so that the distance $\tilde{d}$ between P and the origin of $\mathcal{F}_f$ is minimum, where the ordinate of $\mathcal{F}_f$ is defined along $\tilde{d}$. In this work, we define the point P at the origin of frame $\mathcal{F}_r$ and consider the frame $\mathcal{F}_f$ —here called $\mathcal{F}'_0$ —located on the estimated row centerline, so that $\mathcal{F}'_0$ ordinate lies along the displacement offset e_y .

considering camera feedback (IR imaging, focus of this work) and additional sensor data if available (to be analyzed in future research). It is expected that the performance of this PBVS approach will be mainly driven by the IR camera frame rate, as well as possible delays caused by image processing and filtering of mistaken offset estimations (if applied). While beneficial, the aforementioned periodic pose update approach indirectly introduces disturbances into the odometry, as the camera effectively redefines the robot’s reference frame. Given that the tracking error $\tilde{q}' = q' - q'_d$ is linear, we can assume an disturbance of order δ in the reference q_d to be an equivalent disturbance only affecting the odometry q' . Hence, the disturbed error expressed as $\tilde{q}' = q' - (q'_d \pm \delta)$, affected only by reference disturbances in the order of δ , remains unchanged if we consider $\tilde{q} = (q' \pm \delta) - q'_d$ as affected only by odometry disturbances in the same order. Since the chained form system equations are function only of the term $\tilde{q}'$, and $y'_d = \phi'_d = 0$, the essential dynamics is preserved. Thus, we can model odometry error disturbances as additive terms in the chained form dynamics, by rewriting Eq. (24) to $\dot{z}' = \tilde{S}(z')u$ accordingly:

$$\begin{cases} \dot{z}'_1 = u_1 + \delta_1, \\ \dot{z}'_2 = u_2 + \delta_2, \\ \dot{z}'_3 = u_1 z_2 - z_1 u_2 + \delta_3, \end{cases} \quad (26)$$

where $\delta_1 = \delta_{o1}$, $\delta_2 = \delta_{o2}$, and $\delta_3 = \zeta + \delta_{o3}$. Here, δ_{o1} , δ_{o2} , δ_{o3} represent the odometry disturbances indirectly caused by the periodic pose updates from the camera. The values of these terms can be derived from Eq. (22) by differentiating z in time, after adding a disturbance in each error term. Thus, by considering $\tilde{x}' = \tilde{x} + \delta_x$, $\tilde{y}' = \tilde{y} + \delta_y$ and $\tilde{\phi}' = \tilde{\phi} + \delta_\phi$, where δ_x , δ_y , δ_ϕ are the odometry update disturbances in each pose error variable, we compute the expressions for the disturbance terms that arise in $\dot{z}'$, as follows:

$$\begin{aligned} \delta_{o1} &= \delta_\phi, \\ \delta_{o2} &= c_\phi \delta_x + s_\phi \delta_y, \\ \delta_{o3} &= 2s_\phi \delta_x - \tilde{\phi} c_\phi \delta_x - \tilde{\phi} s_\phi \delta_y + 2c_\phi \delta_y - \delta_\phi c_\phi \tilde{x} \\ &\quad - \delta_\phi s_\phi \tilde{y} - 2x_{icr} \delta_\phi. \end{aligned} \quad (27)$$

B. SLIDING-MODE CONTROL ROBUST TO ODOMETRY DISTURBANCES

One straightforward approach for solving the SSMR control problem would be to consider the robot as a ‘line follower’ system, where the vehicle’s linear speed is fixed to the desired value, with $v_x = v_{dx}$, and its angular speed varies according to the vehicle displacement along the y-axis from the row centerline, with $\omega = -k_\omega (\phi + \gamma y)$, where $k_\omega > 0$ is a tunable control gain, and $\gamma > 0$ defines the ratio between the robot’s heading and the robot’s displacement relative to the row centerline. For example, choosing $\gamma = \frac{\pi/4}{0.2}$ indicates that the robot will be controlled to $\phi = +\pi/4$ rad when $y = -0.2$ m. However, the line follower control (LFC) approach can be dangerous in practice, since the robot will keep a forward

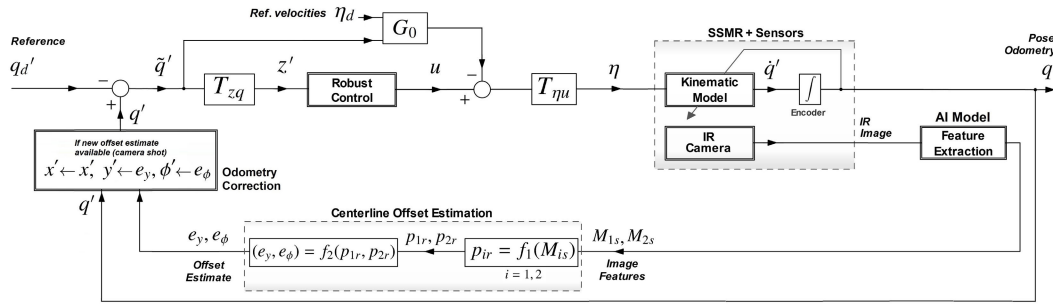


FIGURE 25. Robust control scheme and PBVS approach - The position y' and orientation ϕ' are updated with period h_c from the robot offset estimated by the CNN Model applied to the IR image shot.

constant speed even upon great disturbances, with a risk of collision with the sugarcane stalks of a tight crop row tunnel.

Therefore, it is convenient to propose a robust control law based on the sliding mode control (SMC) approach and the chained form system devised in the previous Sections. The classic unit vector control (UVC) is modified to integrate a chattering attenuation factor into the sign function ('sgn'), which enhances the system performance. Then, the robust control law $u = (u_1, u_2)$ is implemented as:

$$\begin{aligned} u_1 &= -k_1 z_1 - \lambda_1 z_2 |s|^\rho \text{sgn}(s), \\ u_2 &= -k_2 z_2 + \lambda_2 z_1 |s|^\rho \text{sgn}(s), \end{aligned} \quad (28)$$

where $k_1, k_2 > 0$ are positive gains to be tuned, $\rho \in (0, 1)$ is the smoothing factor, $\lambda_1, \lambda_2 > 0$ are positive constants to be computed, and s denotes the sliding surface. We can demonstrate that this control law ensures convergence of the system trajectories $z = [z_1 \ z_2 \ z_3]^T$ to zero if we select a sliding surface $s(z) = z_3$ which attracts all system trajectories and leads them to zero when s goes to zero. Now, we can state the following theorem:

Theorem 1: Consider the tracking control problem for a skid-steering mobile robot with a kinematic model described by Eq. (20). Under the coordinates and control variables transformation matrices, denoted by Eq. (22) and Eq. (23), we obtain the optimal chained form system given by Eq. (26). Assuming that the drift term ζ —denoted by Eq. (24)—and odometry updates $\delta_{o1}, \delta_{o2}, \delta_{o3}$ —denoted by Eq. (27)—are bounded, and using the robust control law given by Eq. (28), the following properties hold: (i) all system signals are bounded; (ii) $\lim_{t \rightarrow \infty} z_1(t), z_2(t) = 0$; (iii) $\lim_{t \rightarrow \infty} s(t) = 0$ and, consequently, $\lim_{t \rightarrow \infty} z_3(t) = 0$; (iv) $\lim_{t \rightarrow \infty} \tilde{q}(t) = 0$.

Proof Stability proof and robustness analysis can be found in Appendix. $\square$

IV. IRNAVSim - SIMULATION ENVIRONMENT FOR THERMAL IMAGING AND ROBOT NAVIGATION

To verify and validate the proposed methodology, comprising a CNN-based perception system combined with an SMC approach, a novel and dedicated 3D computing environment—named IRNavSim—was designed. IRNavSim is meant to simulate not only the robot navigation in a sugarcane crop row, but also to emulate the crop heating

pattern and the IR thermal images collected from a virtual conventional camera.

A. SUGARCANE PRODUCTION FIELD 3D ENVIRONMENT

Firstly, a sugarcane field environment (Fig. 26) was developed in Matlab software to replicate the most critical parameters found in typical row crops in Umoe BioEnergy production field [38], mainly the cane seeding points—as mentioned in Section II to be preset by GNSS-given coordinates. In the designed environment, these features were implemented as a corridor limited by two virtual parallel lines, along which cylindrical cane stalks were positioned above the seeding points and generated as follows (see Fig. 27a):

- (i) “Main Seeding Points” (MSP) randomly spaced between themselves by clearance of $x_{cl} \sim \mathcal{N}(\mu = 0.40 \text{ m}, \sigma = 0.01 \text{ m})$ along the corridor (x), and with clearance of $y_{cl} \sim \mathcal{N}(\mu = 1.50 \text{ m}, \sigma = 0.01 \text{ m})$ across the corridor (y).
- (ii) Five “Surrounding Seeding Points” (SSP) distributed around each MSP along a circle of variable radius $r_{ssp} \sim \mathcal{N}(\mu = 0.25 \text{ m}, \sigma = 0.02 \text{ m})$ and separated by variable $\text{arc}_{ssp} \sim \mathcal{N}(\mu = 72 \text{ deg}, \sigma = 1.5 \text{ deg})$. The number of SSPs around each MSP can vary to more or less than five, with $\sigma = 0.5$. Subsequently, a cylindrical cane stalk is placed above each generated seeding point, with base radius of $r_{cane} \sim \mathcal{N}(\mu = 0.02 \text{ m}, \sigma = 0.005 \text{ m})$ and height $h_{cane} \sim \mathcal{N}(\mu = 2 \text{ m}, \sigma = 0.2 \text{ m})$.

Different corridor lengths and shapes can be selected for this virtual environment to simulate more realistic crop rows such as the ones found in Umoe BioEnergy production field. In this work, a maximum of 50 m length can be selected, and the corridor shape can be either a straight line ($y_d = 0$) or a curved path generated by a composite sine-cosine function of x , being $y = C_1 \sin(C_2 x) \cos(C_3 x)$, where $C_1, C_2, C_3 \in \mathbb{R}^+$ are tunable parameters. Finally, the 3D model of TIBA robot—including its chassis, wheels, and IR thermal camera—was created respecting the dimensions of the real prototype and the camera (Fig. 27b).

B. VIRTUAL THERMAL MAP

A novel virtual environment inside IRNavSim—named *thermal map*—was created to simulate the accumulated heat

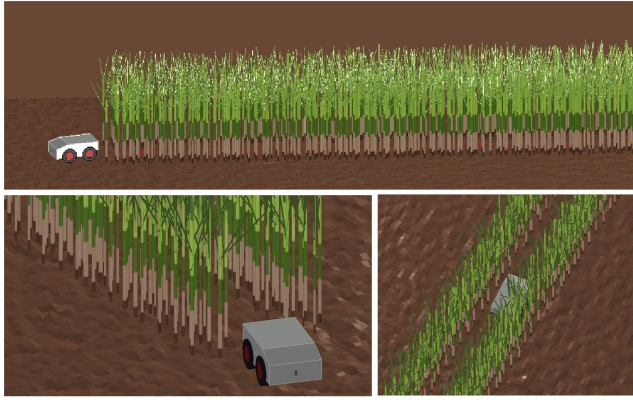


FIGURE 26. IRNavSim- Sugarcane crop in a 3D virtual environment from different perspectives considering a straight corridor.

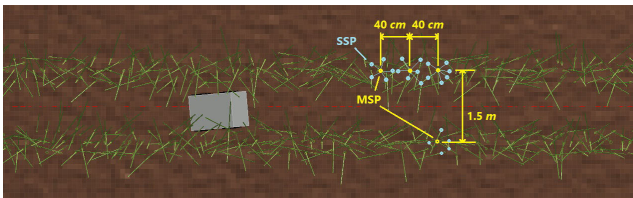


FIGURE 27. IRNavSim - Sugarcane crop in a 3D virtual environment. MSPs in yellow, spaced by $x_d = 0.40$ m and $y_d = 1.5$ m, and respective SSPs in blue, being 5 SSPs around the leftmost MSP and 6 SSPs around the other represented MSPs. The mobile object is a 3D model of the actual robot prototype, with a total length of 1.2 m, distance between wheels center of 0.58 m, wheel radius of $r = 0.20$ m, distance between the center of back wheel to robot rear of 0.26 m, robot height of 0.527 m, and robot width of $2d = 0.72$ m.

in the surfaces that compose the sugarcane environment (i.e., soil, stalks, and plants), and consequently, the temperature matrix that is captured and interpreted by the IR sensor as a “thermal image”. This simulation environment was specially created to validate the crop row following application in sugarcane scenarios. To the best of the authors’ knowledge, no other similar virtual environments or models for thermal imaging were found in the robotics literature or community.

The thermal map environment was constructed from the previously generated sugarcane environment (Section IV-A). Firstly, the ground was divided into a grid of 0.05 m cells with a temperature value assigned to each one, starting from 70°C, except for the cells surrounding each created cane stalk, which started from a lower temperature (Fig. 28), representing water accumulation in stalk bases. Then, a pattern of five leaf branches arranged in a five-pointed star was assigned to each created cane. A specific amount of temperature (1.75°C) is subtracted for each cell that is intercepted by the branches, which aims to simulate temperature drop from the shadow created by the leaves on the soil.

Notice that leaf branches crossing the same cells multiple times will provide multiple temperature subtractions. A low temperature was assigned to the background scenario, which represents low or no radiation from very distant objects. Finally, a gradient color scheme (purple-blue-yellow) was assigned to each grid cell (Fig. 28b), by using the same color scale chosen for capturing the real IR images in field

experiments with TIBA robot. Fig. 29 shows the thermal map environment ground for different corridor lengths and patterns.

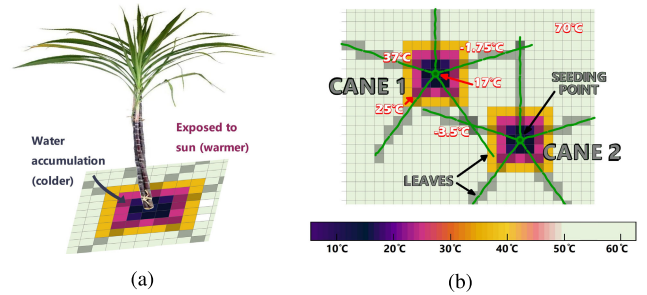


FIGURE 28. IRNavSim Virtual thermal map - (a) heat gradient around sugarcane base, caused by water accumulation and shadow; (b) temperature decrements from $\approx 72^\circ$ spaced leaf branches. For illustrative purposes, the distance between seeding points is depicted as more exaggerated than the reality.

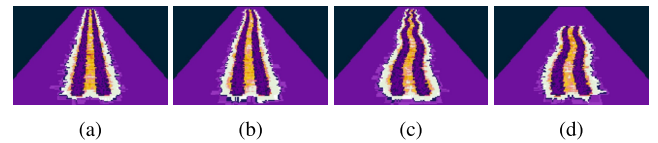


FIGURE 29. IRNavSim Thermal map environment seen from above, with different lengths and patterns - (a) straight long 50 m corridor, $y = 0$; (b) smooth sinusoidal corridor $y = 0.2 \sin(x\pi/20)$; (c) sine-cosine wave $y = 0.5 \sin(0.25x) \cos(0.25x)$, simulating a real corridor; (d) short 20 m corridor. For clarity purposes, cane stalks and leaves are not represented here.

C. VIRTUAL THERMAL CAMERA

Using the created simulation environment described above, a specific function was developed to simulate the IR images in IRNavSim, based on the camera’s extrinsic and intrinsic parameters as defined in Section II-C3. In this work, these camera parameters are all customizable. However, during the simulation of the robot movement, to represent disturbances in the camera assembly, the camera orientation angles and its position coordinates, denoted by $\theta = (\theta_r, \theta_p, \theta_y)$ and $c = (c_x, c_y, c_z)$, vary according to a normal distribution with mean and standard deviation given respectively by: $\mu_\theta = (0, 0.2618, 0)$ rad and $\sigma_\theta = 3.4E-3$ rad; $\mu_c = (0.65, 0, 0.22)$ m and $\sigma_c = 2.0E-3$ m.

In addition, random patterns of trapping leaves were added to the upper half of the image, representing the noise from the leaf branches falling inwards the crop row tunnel. Forty different “noise” patterns were extracted from real IR images collected from the field, where each of them was randomly selected according to the robot’s position x along the corridor. Fig. 30 shows virtual IR images for different camera parameters by utilizing the previous simulation environment and the exact resolution and original focus length of the thermal camera used during the field experiments (FLIR[®] One V2): pixel resolution $\beta_x \times \beta_y$ of 120×160 pixels, $f = 2.3 \mu\text{m}$, $\rho = \rho_h = \rho_w = 12 \mu\text{m}$, as per the datasheet, resulting in $\alpha_x = 1440 \mu\text{m}$ and $\alpha_y = 1920 \mu\text{m}$.

The thermal map environment was created around the daytime (6:00 PM) in which the IR images were collected

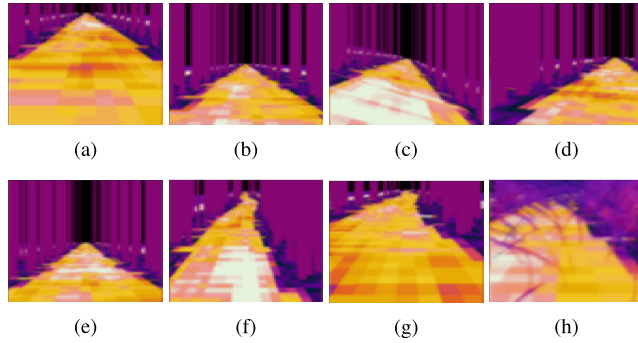


FIGURE 30. IRNavSim Virtual IR image with different camera parameters - (a) default setting, camera position coordinates $c_x = 0.65$ m, $c_y = 0$ m, $c_z = 0.22$ m, rotation only downwards by 15 deg around y_r -axis; (b) rotation around y_r -axis by 0 deg; (c) rotation around x_r -axis by -15 deg; (d) rotation around z_r -axis by 15 deg; (e) increased focus length to $f = 0.04$ m; (f) image for a curved row; (g) increased camera height to $c_z = 0.50$ m for the same curved row; (h) noise included, trapping leaf branches, with Gaussian blur effect added, aiming to simulate the image quality loss observed for the real thermal camera utilized in the field.

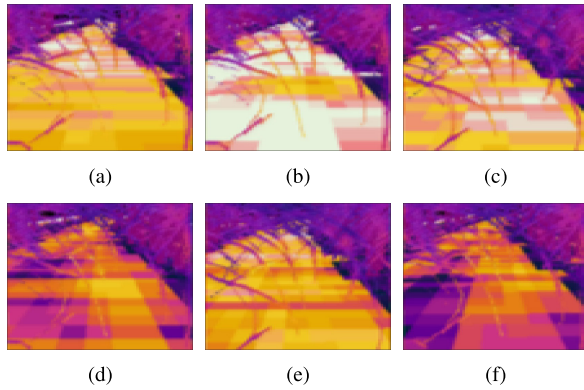
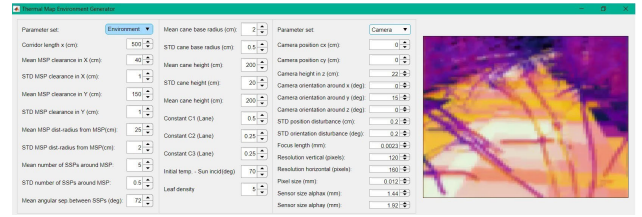
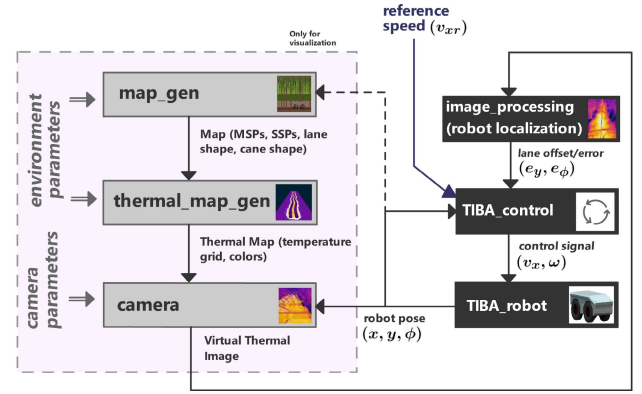


FIGURE 31. IRNavSim Virtual IR image with different environmental and field parameters to emulate distinct thermal patterns - (a) default environment, 6:00 PM (70 deg sun incidence), leaf density of 5 branches; (b) earlier, 5:00 PM (80 deg sun incidence); (c) earlier, 5:30 PM (75 deg sun incidence); (d) later, 7:00 PM (60 deg sun incidence); (e) slightly taller stalks, leaf density of 6 branches; (f) taller stalks, leaf density of 7 branches.

during the experiment at Umoe BioEnergy production field. Although not utilized in this work, small variations can be considered to represent earlier or later day periods—by changing the initial cell temperature from 70°C—as well as to represent shorter or taller stalks—by changing the leaf density from 5 branches around each seeding point. Fig. 31 shows how the sugarcane virtual environment is captured by the virtual camera by considering different variations in the aforementioned environmental and field parameters. A graphical user interface (Fig. 32a), named Thermal Map Environment Generator, was created to allow the user to configure all the possible parameters for the environment and IR camera. Fig. 32b depicts a flowchart that illustrates how the created functions interact to finally generate the virtual thermal image, and how this environment is integrated into the control and image processing functions that can be created for the crop row following task with TIBA robot.



(a)



(b)

FIGURE 32. IRNavSim setup and framework - (a) thermal map environment generator and its configuration parameters; (b) control and image processing functions flowchart.

TABLE 4. Summary of simulations with IR image feedback - For all cases: simulation time of 30 s, integration period of 0.001 s, sinusoidal corridor shape function $y = 0.5 \sin(0.25x) \cos(0.25x)$ in m, initial pose $(x_0, y_0, \phi_0) = (0.40, 0.05, 0.00)$ in (m,m,rad), and target velocity $v_{dx} = 0.60 \text{ ms}^{-1}$

Sim. ID	Method	Cam. update period h_c (s)	Control gains and parameters
#1	Baseline +IR	0.1	$k_1 = k_2 = 2$, $\alpha_0 = \alpha_1 = 2$, $\epsilon_1 = 0.1$, $z_{d0} = [0.5 \ 0.5]^T$
#2	SMC +IR	0.1	$k_1 = k_2 = 3.5$, $\lambda_1 = \lambda_2 = 5$, $\rho = 0.3$
#3		0.05	
#4		1	

It is worth noting that IRNavSim allows for varying the thermal profile by adjusting the intensity and angle of solar radiation incidence. For the time being, these parameters are directly adjusted at the GUI instead of as a function of daytime and season. These limitations could be addressed as part of future research.

V. ROW FOLLOWING IN SIMULATION ENVIRONMENT

To verify the row crop following methodology proposed in this work (Fig. 25)—which considers IR camera image processing for periodic updates of the tracking reference closing the sliding mode control loop (SMC+IR)—a set of simulations was carried out on IRNavSim.

Firstly, we compare the application of the proposed SMC approach with other approaches, such as the line follower

TABLE 5. Performance metrics for simulations with LFC, Baseline and SMC with IR camera feedback (maximum tracking error, RMSE, MAD, offset estimation μ , σ). *LFC approach provides zero tracking error for linear speed, as it is permanently fixed.(a) LFC vs. Baseline vs. SMC (camera update period $h_c = 0.1$ s).

MAE	LFC	Baseline	SMC
$\tilde{y}'$ (m)	0.338	0.064	0.027
$\tilde{\phi}'$ (rad)	0.500	0.148	0.021
RMSE	LFC	Baseline	SMC
$\tilde{y}'$ (m)	0.087	0.030	0.016
$\tilde{\phi}'$ (rad)	0.105	0.051	0.018
MAD	LFC	Baseline	SMC
v_x (m s ⁻¹)	0.000*	0.183	0.004
ω (rad s ⁻¹)	0.148	0.233	0.031
ω_L, ω_R (rad s ⁻¹)	0.267	1.015	0.063
Offset estimation	LFC	Baseline	SMC
$\mu_{ \tilde{y}'-e_y }$ (m)	0.060	0.030	0.028
$\sigma_{ \tilde{y}'-e_y }$ (m)	0.057	0.020	0.020
$\mu_{ \tilde{\phi}'-e_\phi }$ (rad)	0.059	0.036	0.017
$\sigma_{ \tilde{\phi}'-e_\phi }$ (rad)	0.097	0.034	0.015

(b) Only SMC (different camera update periods h_c).

MAE	$h_c = 0.05$ s	$h_c = 0.1$ s	$h_c = 1$ s
$\tilde{y}'$ (m)	0.026	0.027	0.173
$\tilde{\phi}'$ (rad)	0.018	0.021	0.125
RMSE	$h_c = 0.05$ s	$h_c = 0.1$ s	$h_c = 1$ s
$\tilde{y}'$ (m)	0.016	0.016	0.096
$\tilde{\phi}'$ (rad)	0.017	0.018	0.063
MAD	$h_c = 0.05$ s	$h_c = 0.1$ s	$h_c = 1$ s
v_x (m s ⁻¹)	0.005	0.004	0.023
ω (rad s ⁻¹)	0.031	0.031	0.047
ω_L, ω_R (rad s ⁻¹)	0.063	0.063	0.156
Offset estimation	$h_c = 0.05$ s	$h_c = 0.1$ s	$h_c = 1$ s
$\mu_{ \tilde{y}'-e_y }$ (m)	0.029	0.028	0.033
$\sigma_{ \tilde{y}'-e_y }$ (m)	0.017	0.020	0.026
$\mu_{ \tilde{\phi}'-e_\phi }$ (rad)	0.016	0.017	0.060
$\sigma_{ \tilde{\phi}'-e_\phi }$ (rad)	0.016	0.015	0.040

control (LFC)—mentioned in Subsection III-B—and a *Baseline controller* represented by the kinematic control solution proposed in the established work from Kozłowski and Pazderski [42]. It is expected that the SMC presents better performance, since, unlike the LFC and the Baseline approaches, it was designed to be robust to the system disturbances, especially these caused by odometry update from offset estimation via IR camera. For this comparison, we consider that the robot tracking errors regarding the row centerline are estimated at every 0.1 s. Finally, we evaluate the proposed SMC+IR approach under varying camera update/frame rates derived from IR cameras available on the market, aiming to investigate how different IR cameras—as discussed along this work to be notably subjected to low frame rates or image processing delays—impact the overall performance of the proposed method. In this work, we present examples of the simulations carried out (Table 4), described through images, graphs and performance metrics. This table also outlines the parameters utilized in the simulations, such as total time, gains, update rates, and corridor shape. It is worth noticing that the LFC+IR method demonstrated unsatisfactory performance in the simulations. For simplicity, only its performance metrics are included in this paper, while the simulation plots are omitted.

For comparative purposes, three performance metrics were also used for assessment of the results, as summarized in Table 5: (i) MAE (maximum absolute error) and (ii) RMSE (root mean square error) of the linear and angular tracking errors, $\tilde{y}$ and $\tilde{\phi}$; (iii) MAD (mean-absolute deviation) of the robot inputs, v_x and ω , and wheel speeds ω_L, ω_R ; (iv) offset estimation performance from IR camera and AI model, represented by the mean and the standard deviation of the absolute differences between real offset $\tilde{y}'$, $\tilde{\phi}'$ and camera estimates e_y, e_ϕ .

Additionally, animations of some of the simulated rides with TIBA robot can be visualized in the accompanying video clip.⁶ It is worth mentioning that the simulations in the IRNavSim virtual environment were exhaustively carried out using the proposed CNN-based centerline estimation method tuned by using IR images collected from real sugarcane fields. However, additional field trials for the closed-loop system with TIBA robot were not conducted yet due to costly logistics and unavailable field sites during harvesting seasons.

For demonstration and validation, the control methods were tested for different target linear velocities and for straight, curved, and sinusoidal shapes, aiming to represent realistic crop rows as usually found in Umoe BioEnergy production field. In the sugarcane plantations of Umoe BioEnergy farm, the rows are not perfectly straight, but smoothly curved mainly due to uneven/inclined terrain and other factors related to nature and field delimitation. In this work, for simplification purposes, we will show only the results obtained from the sinusoidal function $y = 0.5 \sin(0.25x) \cos(0.25x)$ expressed in the world frame $\mathcal{F}_0$, with distance between cane stalks of $d_{row} = 1.5$ m, initial robot pose of $x_0 = 0.40$ m, $y_0 = -0.05$ m, $\phi_0 = 0$ rad, and reference linear velocity of $v_{dx} = 0.60$ m s⁻¹.

Notice that this curve represents the worst-case scenario due to having a minimum radius of curvature significantly smaller than the range of radius of curvatures R commonly found in Umoe BioEnergy farm, that is, 125-1000 m as mentioned in Section II-B2 and illustrated in Fig. 33. The curve used for Simulations #1-3 presents a minimum $R = 16$ m, sharply accentuated in relation to the minimum radius of curvature found in the sugarcane field, considering that it is

⁶Please, see the accompanying video clip available at <https://youtu.be/sz7cq4BDtJM> (long version) or <https://youtu.be/qR1QuiquKq4> (short version).

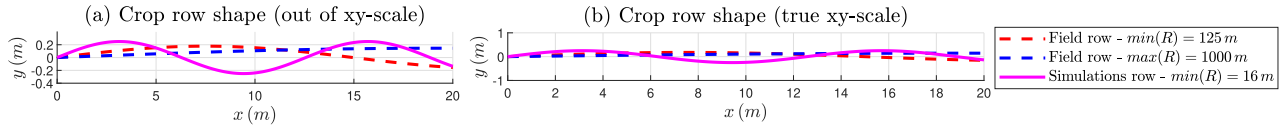


FIGURE 33. Comparison of crop row formats between those commonly found in the field and simulated ones. As seen from the top view, in the xy -plane - (a) augmented view with x and y axes not to scale; (b) true view with x and y axes in unit scale.

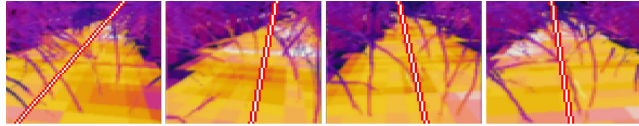


FIGURE 34. Screenshots of virtual IR camera in IRNavSim with fitted centerline during the SMC+IR approach simulation along the crop row.

calculated using the formula $R(x) = \left| (1 + [y'(x)]^2)^{\frac{3}{2}} / y''(x) \right|$, where the term y' means the first derivative of y with respect to x , and so on. For all simulations, the virtual IR camera images—along the sinusoidal crop row in the IRNavSim environment—also show the estimated centerline at every camera shot, as per the example screenshots shown in Fig. 34.

A. SIMULATIONS WITH IR IMAGE FEEDBACK - COMPARING BASELINE+IR AND SMC+IR (CAMERA UPDATE PERIOD 0.1 s)

In this Section, we detail the simulation results from two different control strategies combined with IR thermal imaging feedback to evaluate their performance and feasibility: the Baseline control from Kozłowski and Pazderski [42], and the sliding mode control. Although several initial conditions and control parameters were tested, in this work we only consider one initial pose and the set of parameters depicted in Table 4, aiming for simplicity, comparison and demonstration purposes. All tested initial orientation angles were set to be in the range of $(-\pi/2, \pi/2)$ rad, since the real case scenario described in this work considers the camera to be mounted always aligned with the sagittal axis of the vehicle. For this comparison, we consider that the tracking errors regarding the row centerline are estimated at every 0.1 s, which is close to the update rate of FLIR[®] One V2—the real IR camera used in field experiments with TIBA robot (this camera provides frame rate of 8.7 Hz, equivalent to a frame update period of ≈ 0.11 s).

1) SIMULATION #1 - BASELINE+IR WITH CAMERA UPDATE PERIOD 0.1 s

The kinematic control proposed by Kozłowski and Pazderski [42] was tested by using many different initial conditions, and, for comparison purposes, the control parameters were selected based on the recommendations and boundaries discusses in that work. Simulation results demonstrated that this Baseline, as expected, encounters difficulties in ensuring robustness to the system and odometry updates disturbances. As we can observe from the example in Fig. 35a and 35b, the

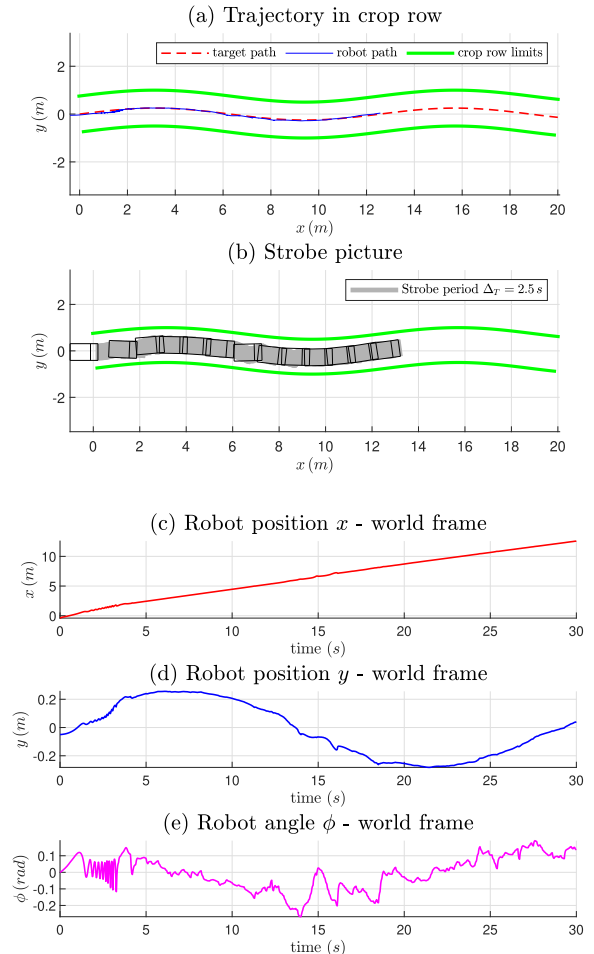


FIGURE 35. Baseline controller [42]+IR Simulation #1 - (a) robot trajectory along sinusoidal crop row $y = 0.5 \sin(0.25x) \cos(0.25x)$ seen from above; (b) stroboscopic picture at every 2.5 s; (c)-(e) time evolution of the robot pose $q(t) = [x \ y \ \phi]^T$ with respect to the world frame. Control parameters: $k_1 = k_2 = \alpha_0 = \alpha_1 = 2$, $\epsilon_1 = 0.1$, and initial $z_d = [0.5 \ -0.5]^T$. Update period: $h_c = 0.1$ s.

system maneuvering is not smooth and presents accentuated skid, as well as fails to track the desired linear velocity, although it was able to ride along the corridor keeping itself reasonably centralized on the corridor. Fig. 35c to 35e show the evolution over time of the robot coordinates $q = (x, y, \phi)$ with respect to the world frame.

Table 5a shows the observed maximum absolute linear and angular tracking error with respect to the corridor centerline, after the system transient period. The maximum linear error represents approximately only 12.2% of the robot width, 5.9% of the row width, and 22.6% of the

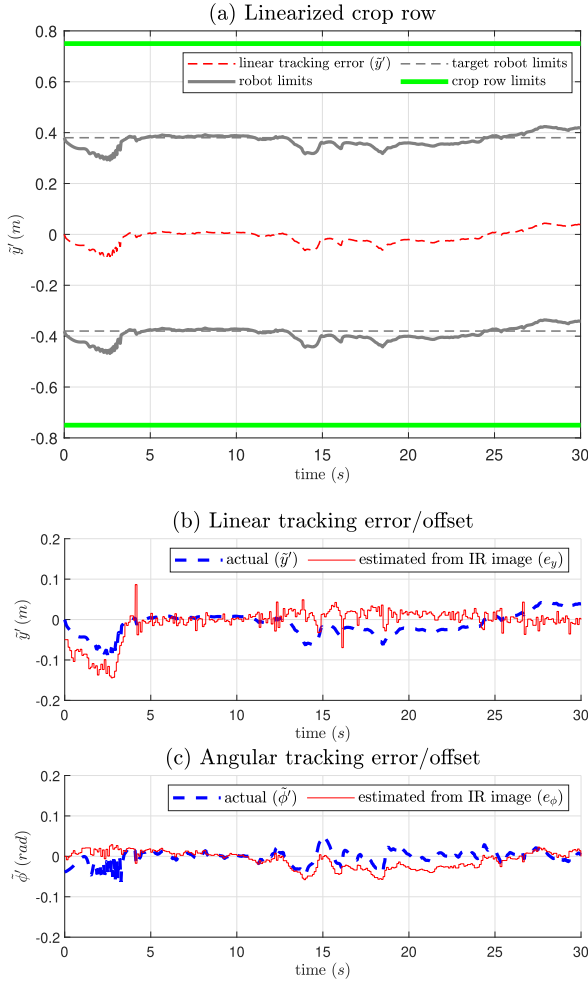


FIGURE 36. Baseline controller [42]+IR Simulation #1 - (a) time evolution of robot linear tracking error ($\tilde{y}'$) and physical chassis limits compared to the limits of the "linearized" crop row; (b)-(c) comparisons between the robot linear/angular offsets (e_y, e_ϕ) estimated from the IR image processing algorithm (update period $h_c = 0.1$ s) and the actual tracking errors ($\tilde{y}', \tilde{\phi}'$) with respect to the row centerline. Control parameters: $k_1 = k_2 = 2, \alpha_0 = \alpha_1 = 2, \epsilon_1 = 0.1$, and initial $z_d = [0.5 \ -0.5]^T$.

maximum admissible distance between the robot lateral and the row crop limits. This indicates a highly superior path tracking performance if compared to the LFC approach. Fig. 36b and 36c compare the actual robot tracking errors to the centerline ($\tilde{y}', \tilde{\phi}'$) with the offsets (e_y, e_ϕ) that were estimated from the IR image processing algorithm presented in Section II-C3. It is worth mentioning that the performance of the offset estimation via IR camera remained consistent, presenting small absolute differences between real offset and camera estimates, as shown in Table 5a. Overall, the offset estimates in the Baseline simulations were slightly better than those in the LFC simulations. This can be attributed to the higher sensitivity and reduced accuracy of estimates near the edges of the camera's field of view, a condition more frequently encountered in LFC due to greater yaw disturbances during robot maneuvers. Fig. 37a and 37b show, respectively, evolution of the variable w and the chained form system z_1, z_2 —as described by [42]—as well as

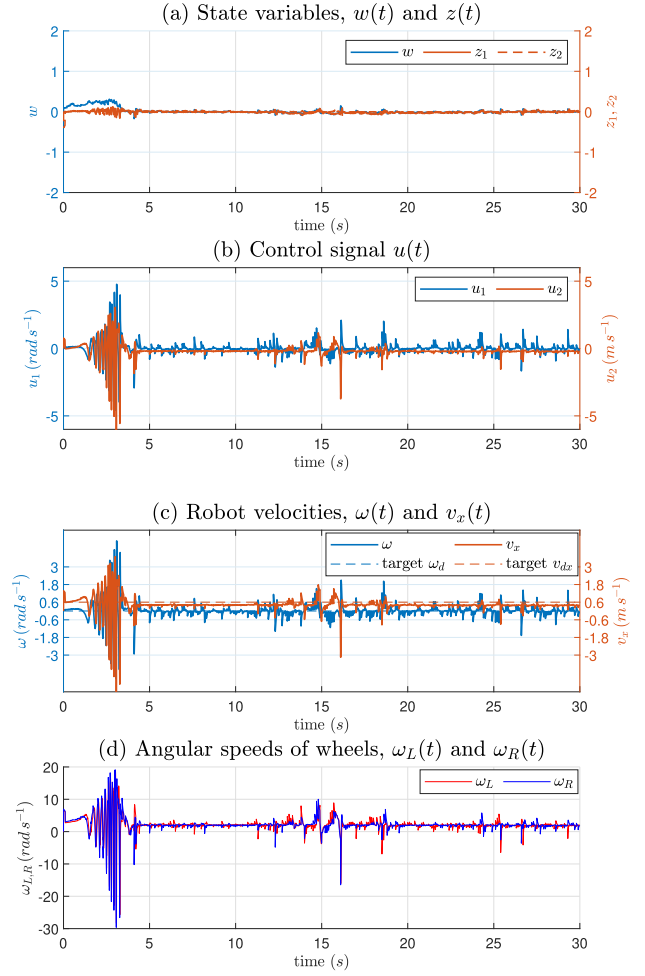


FIGURE 37. Baseline controller [42]+IR Simulation #1 - (a) time evolution of the variable $w(t)$ and the chained form system variables, $z_1(t), z_2(t)$; (b) time evolution of the control signals $u_1(t)$ and $u_2(t)$; (c) time evolution of the driving and steering velocities $\eta(t) = [v_x \ \omega]^T$ compared to the reference velocities $v_{dx} = 0.60 \text{ m s}^{-1}$ and $\omega_d = 0 \text{ rad s}^{-1}$; (d) time evolution of the angular speeds of the left and right wheels, $\omega_L(t)$ and $\omega_R(t)$, respectively. Control parameters: $k_1 = k_2 = \alpha_0 = \alpha_1 = 2, \epsilon_1 = 0.1$, and initial $z_d = [0.5 \ -0.5]^T$. Update period: $h_c = 0.1$ s.

the velocity control signals, and the angular speed of the robot wheels. Fig. 37c shows, as expected, that the system fixes a constant linear velocity of $v_{dx} = 0.60 \text{ m s}^{-1}$, and attempts to keep the null reference angular velocity ω_d by applying differential speeds (Fig. 37d) to allow for the required curves (performance metrics in MAD shown in Table 5a). As expected, the controller proposed by [42] was not assessed for robustness to odometry disturbances, which can be observed from the high-frequency noise of the robot control signals and wheel speeds, and from eventual saturation of the system actuators.

2) SIMULATION #2 - SLIDING MODE CONTROL (SMC+IR) WITH CAMERA UPDATE PERIOD 0.1 s

In this simulation, we consider the proposed method in this work: SMC with feedback from IR thermal imaging for offset estimation. This approach was tested by using many different

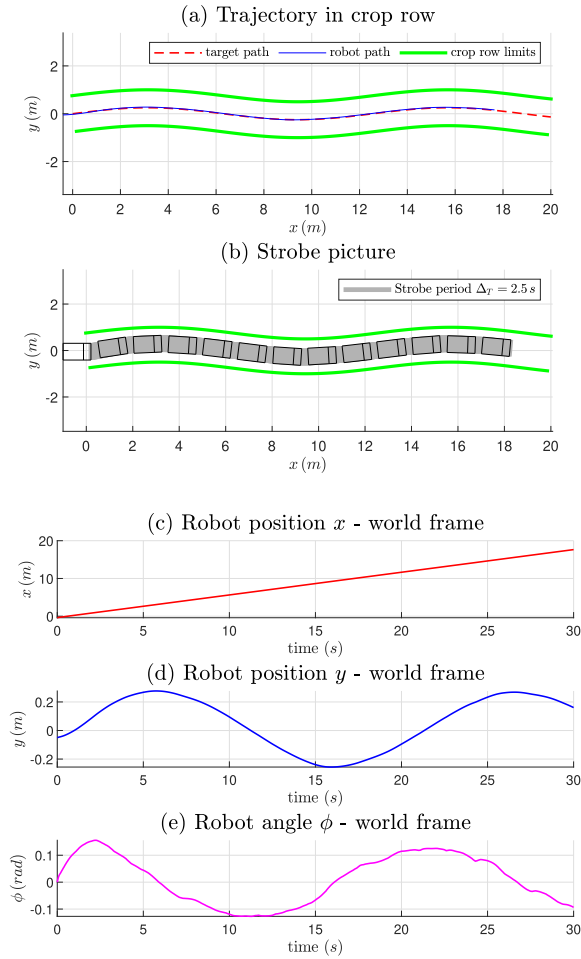


FIGURE 38. SMC+IR Simulation #2 - (a) robot trajectory along sinusoidal crop row $y = 0.5 \sin(0.25x) \cos(0.25x)$ seen from above; (b) stroboscopic picture at every 2.5 s; (c)-(e) time evolution of the robot pose $q(t) = [x \ y \ \phi]^T$ with respect to the world frame. Control parameters: $k_1 = k_2 = 3.5$, $\lambda_1 = \lambda_2 = 5$, $\rho = 0.3$. Update period: $h_c = 0.1$ s.

initial conditions and control parameters, aiming to evaluate tracking performance and robustness. We can notice from the example depicted by Fig. 38a and 38b that the robot never collided with the crop and was able to ride along the corridor keeping itself reasonably centralized on the corridor, having its mechanical limits as a relatively large clearance to the crop walls.

Table 5a shows the observed maximum absolute linear and angular tracking error with respect to the corridor centerline, after the system transient period. The maximum linear error represents approximately only 3.8% of the robot width, 1.8% of the row width, and 6.9% of the maximum admissible distance between the robot lateral and the row crop limits. Analysis of the results reveals that the proposed SMC law not only ensured that the robot achieved the pose tracking goal but also provided robustness against system and odometry update disturbances, significantly outperforming the LFC and Baseline approaches under the same conditions.

Fig. 39b and 39c compare the actual robot tracking errors to the centerline ($\tilde{y}'$, $\tilde{\phi}'$) with the offsets (e_y , e_ϕ) that were

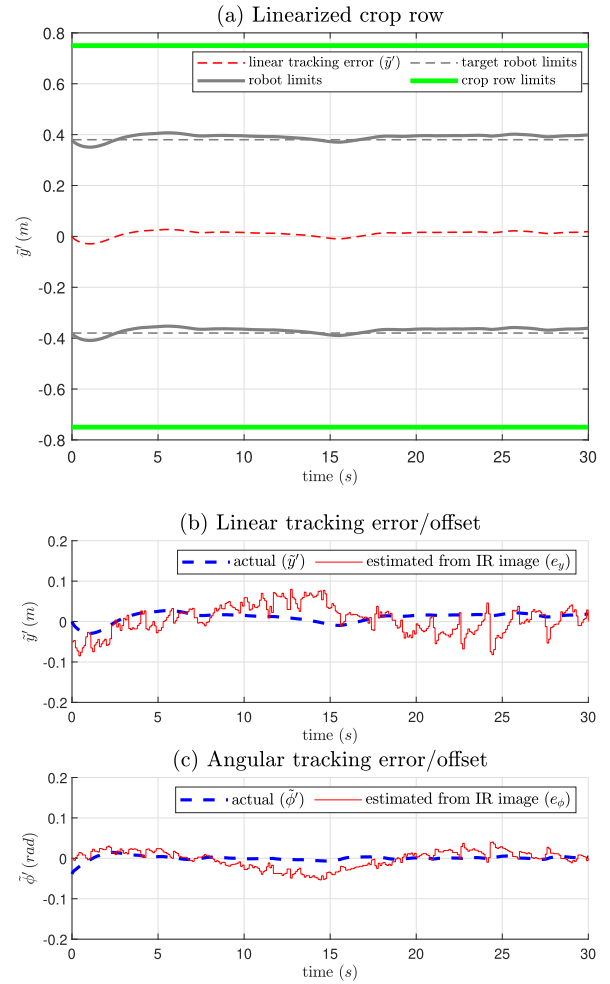


FIGURE 39. SMC+IR Simulation #2 - (a) time evolution of the robot linear tracking error ($\tilde{y}'$) and physical chassis limits compared to the limits of the “linearized” crop row; (b)-(c) comparisons between the robot linear/angular offsets (e_y , e_ϕ) estimated from IR image processing algorithm (update period $h_c = 0.1$ s) and the actual tracking errors ($\tilde{y}'$, $\tilde{\phi}'$) with respect to the row centerline. Control parameters: $k_1 = k_2 = 3.5$, $\lambda_1 = \lambda_2 = 5$, $\rho = 0.3$.

estimated from the IR image processing algorithm presented in Section II-C3. We can verify that the performance of the offset estimation via IR camera remained consistent, presenting significantly small absolute differences between real offset and camera estimates (as per Table 5a). We can verify that the overall mean of the offset estimates in the SMC simulations were similar to those in the Baseline simulations, however it remained more consistent (smaller σ) if compared to all methods. This can be attributed to the row centerline remaining predominantly centralized within the camera’s field of view for most of the time by using SMC, with the angular offset consistently staying close to zero, avoiding the screen edges where higher sensitivity and reduced estimate accuracy are more pronounced.

Fig. 40a to 40c show respectively the sliding mode control (SMC) variables, the velocity control signals, and the angular speed of the robot wheels. Fig. 40c shows that the system

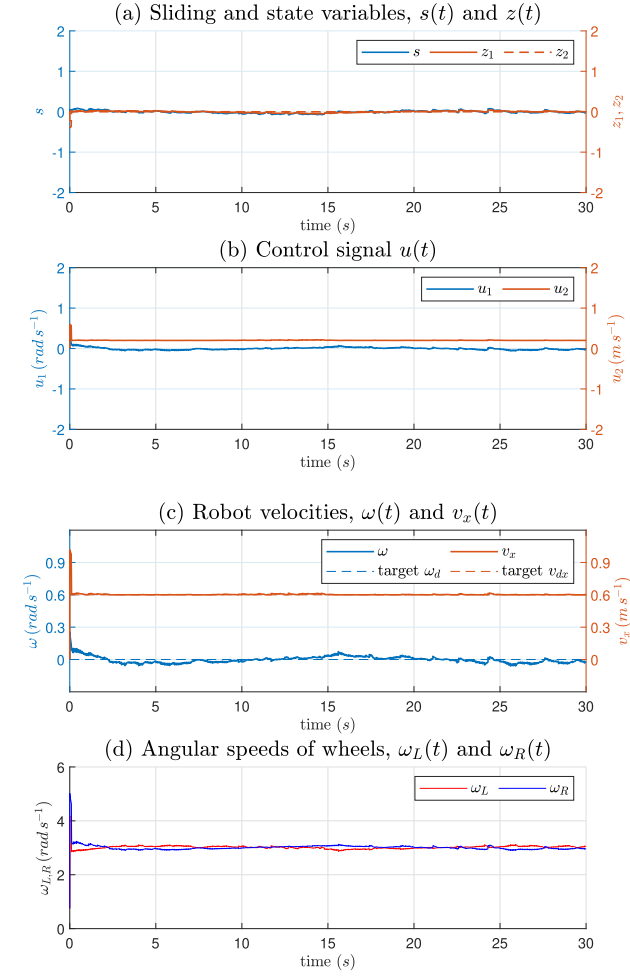


FIGURE 40. SMC+IR Simulation #2 - (a) time evolution of the sliding variable $s(t)$ and the chained form system variables, $z_1(t)$ and $z_2(t)$; (b) time evolution of the control signals $u_1(t)$ and $u_2(t)$; (c) time evolution of the driving and steering velocities $\eta(t) = [v_x \ \omega]^T$ compared to the reference velocities $v_{dx} = 0.60 \text{ m s}^{-1}$ and $\omega_d = 0 \text{ rad s}^{-1}$; (d) time evolution of the angular speeds of the left and right wheels, $\omega_L(t)$ and $\omega_R(t)$, respectively. Control parameters: $k_1 = k_2 = 3.5$, $\lambda_1 = \lambda_2 = 5$, $\rho = 0.3$. Update period: $h_c = 0.1 \text{ s}$.

satisfactorily succeeds in controlling the robot to its reference linear velocity of $v_{dx} = 0.60 \text{ m s}^{-1}$ and null reference angular velocity ω_d where we can see from Fig. 40d) the small differential speeds being applied to allow for the required smooth curves (performance metrics in MAD shown in Table 5a). Once again, it is evident that the proposed SMC law enabled the robot to successfully track the linear velocity goal while maintaining a satisfactorily small variation in angular velocity, in contrast to the results observed with the LFC and Baseline approaches under the same conditions.

B. SIMULATIONS #3 AND #4 - SLIDING MODE CONTROL (SMC+IR) VARYING CAMERA UPDATE PERIODS TO 0.05 s AND 1 s

Finally, in this Subsection, we investigate how different odometry update periods, determined by varying IR camera frame rates, may affect the performance of the proposed

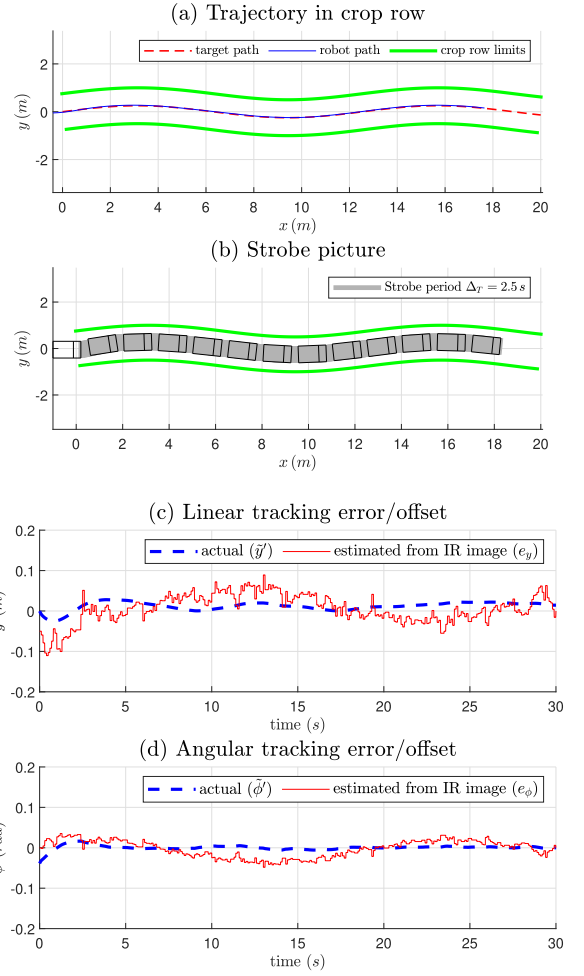


FIGURE 41. SMC+IR Simulation #3 - (a) robot trajectory along sinusoidal crop row $y = 0.5 \sin(0.25x) \cos(0.25x)$ seen from above; (b) stroboscopic picture at every 2.5 s; (c)-(d) comparisons between the robot linear/angular offsets (e_y , e_ϕ) estimated from IR image processing algorithm (update period $h_c = 0.05 \text{ s}$) and the actual tracking errors (y' , ϕ') with respect to row centerline. Control parameters: $k_1 = k_2 = 3.5$, $\lambda_1 = \lambda_2 = 5$, $\rho = 0.3$.

SMC+IR approach. For this analysis, we consider a faster update period of 0.05 s, representing a top-tier IR sensor—such as the FLIR[®] Lepton camera (which features frame rate of 30 Hz, equivalent to frame update period of $\approx 0.03 \text{ s}$). Additionally, we assess a worst-case scenario characterized by significant image processing delays or IR cameras featuring very modest frame rate. Here, we represent this challenging case as if the camera has an update period of 1 s. For simplicity purposes, this Subsection will only show the plots of the 2-D trajectory and comparison between offset estimates and real tracking errors. The performance of the control signals can be evaluated according to the MAD metrics shown in Table 5b. From simulation results, we verify from Fig. 41a and 41b ($h_c = 0.05 \text{ s}$) and Fig. 42a and 42b ($h_c = 1 \text{ s}$) that the robot is still capable to drive centralized on the crop row. As expected, we observe significantly higher offset for the $h_c = 1 \text{ s}$ case, which can attributed to the offset estimation delay that also propagates to a delayed control

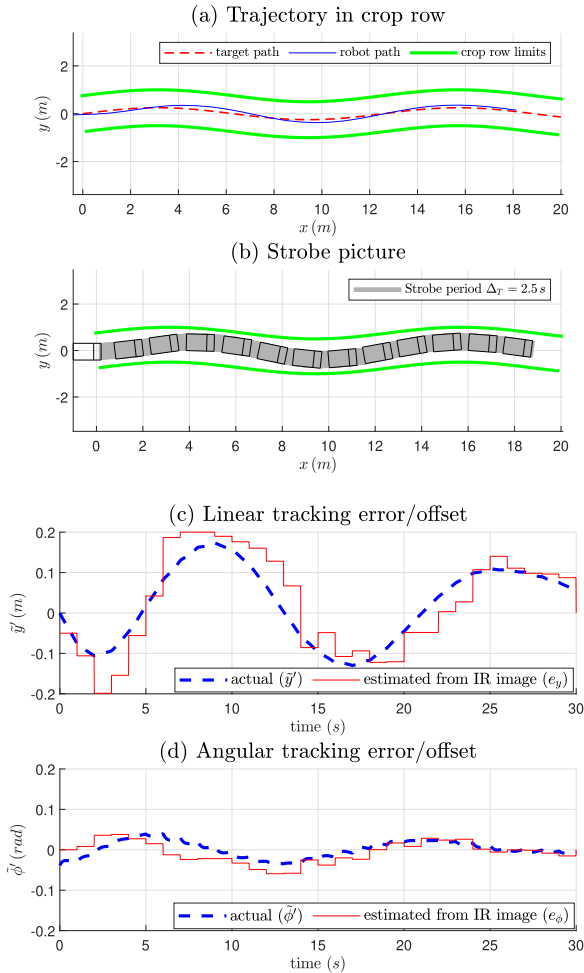


FIGURE 42. SMC+IR Simulation #4 - (a) robot trajectory along sinusoidal crop row $y = 0.5 \sin(0.25x) \cos(0.25x)$ seen from above; (b) stroboscopic picture at every 2.5 s; (c)-(d) comparisons between the robot linear/angular offsets (e_y, e_ϕ) estimated from IR image processing algorithm (update period $h_c = 1$ s) and the actual tracking errors ($\tilde{y}', \tilde{\phi}'$) with respect to the row centerline. Control parameters: $k_1 = k_2 = 3.5$, $\lambda_1 = \lambda_2 = 5$, $\rho = 0.3$.

action from the robot. For both cases, Table 5b shows the maximum absolute linear/angular tracking error with respect to the corridor centerline after the system transient period, as well as the RMSE and MAD performance metrics. We can verify that the system performance did not change when a smaller update period was considered ($h_c = 0.05$ s case), implying that even more sophisticated cameras with faster frame rates may not be required for this application. However, it can also be observed that the delay caused by the large update period ($h_c = 1$ s case) negatively impacted the system's trajectory tracking performance, making it comparable to or worse than the Baseline. Nevertheless, the control signals remained better and smoother than those of the Baseline. Fig. 41c and 41d ($h_c = 0.05$ s) and Fig. 42c and 42d ($h_c = 1$ s) compare the actual robot tracking errors ($\tilde{y}', \tilde{\phi}'$) with the offsets (e_y, e_ϕ) that were estimated from the IR image processing algorithm presented in Section II-C3. The performance of the offset estimation via IR camera (as per

Table 5b) remained consistent with small absolute differences between real offset and camera estimates, although we clearly observe the estimate delay for the $h_c = 1$ s case, as expected. We can clearly verify that the overall mean of the offset estimates in the SMC simulations were similar to those in the Baseline simulations, however it remained more consistent (smaller σ) if compared to all methods. This can be attributed to the row centerline remaining predominantly centralized within the camera's field of view for most of the time by using SMC, with the angular offset consistently staying close to zero, avoiding the screen edges where higher sensitivity and reduced estimate accuracy are more pronounced.

VI. CONCLUSION AND FUTURE RESEARCH

This work presents a novel crop row-following methodology based on IR thermal imaging, comprising a deep learning-based perception method and a robust control algorithm for mobile robot operation in sugarcane fields. It also includes the kinematic modeling, robustness analysis, simulation results, and field experiments for a skid-steering mobile robot, named TIBA (Tankette for Intelligent BioEnergy Agriculture [7]). A 3D simulation environment for emulating different thermal image patterns and crop field conditions, named IRNavSim, was developed in Matlab to verify and validate the crop row following methodology. Fieldwork in sugarcane production and experimental fields was conducted to collect multisensor data and test the robot's mechanical structure in uneven and rough terrain. A case study was presented to compare a few mobile robot control approaches: the Baseline controller proposed by Kozłowski and Pazderski [42], the line follower controller (LFC) and the sliding mode controller (SMC), wherein the latter showed better performance metrics than its counterparts, as shown in Table 5. In the proposed position-based visual servoing (PBVS) scheme, an IR thermal imaging system based on a Convolutional Neural Network (CNN) model has been synergized with the SMC approach to enable accurate centerline offset estimation and robot motion with different camera update rates in a crop field with a sinusoidal corridor. The novel combined perception and control strategy demonstrates a promising efficiency for solving the crop row following problem and lays the foundation for future research and implementation in all phases of mobile robot navigation. The use of infrared images for crop feature detection, providing the robot offset estimation from the row centerline, offers higher reliability and lower computational cost than RGB images, as illustrated in Fig. 21. It can also be observed in Table 3 that the CNN model achieved superior performance when trained with IR images, with an accuracy of 84.3%, a precision of 86.5%, and an F1-score of 0.84. Despite the need for large training datasets, CNN models for processing thermal IR images show relevant potential in addressing various field conditions in sugarcane fields. The affordability and portability of the onboard processing unit and IR camera required for the

crop row following, such as a Raspberry Pi integrated with a low-cost embedded FLIR[®] Lepton camera, make it an attractive solution for robot navigation in sugarcane fields. In conclusion, TIBA robot and the proposed crop row following methodology represent a significant advancement in agricultural robotics, with the potential to enhance precision agriculture in sugarcane fields and other grass-type crop production.

Future research will address the complete navigation problem, including autonomous maneuvers for row switching in the headlands, use of multiple IR cameras for leaf occlusion avoidance and 3D point cloud generation of the crop row, and multi-sensor fusion with other low-cost sensor types, such as sun sensors, which can provide valuable data for sunlight-based guidance when the robot leaves the sugarcane tunnel. Although there is already some diversity in the dataset, as shown in Table 1, the next steps will also involve collecting more data from sugarcane fields under various field conditions, such as different daylight, weather, plant heights, and seasons to better understand the environment's thermal profile and improve the training of the CNN model and the IRNavSim environment. A more comprehensive mapping of the thermal profile will enable the identification of optimal combinations of time and weather for data collection, ultimately enhancing the accuracy and reliability of the corridor feature detection, and consequently, the centerline estimation. Finally, a future work will consider the implementation of a robust image-based visual servoing (rIBVS) approach, enabling the control design to be implemented in the image frame rather than in Cartesian frame, thereby coping with the uncertainties related to camera extrinsic and intrinsic parameters [55], as well as external disturbances from uneven and rough terrain.

APPENDIX SLIDING MODE CONTROL STABILITY PROOF AND ROBUSTNESS ANALYSIS

To verify the stability of the closed loop system using the robust control law proposed in Subsection III-B, we consider two Lyapunov functions: $V_z = z_1^2 + z_2^2$, to analyze the convergence of z_1, z_2 to zero, and $V_s = s^2 = z_3^2$ to ensure $z_3 = s \rightarrow 0$ after $z_1, z_2 \rightarrow 0$. Firstly, let us analyze V_z time-derivative, and replace Eq. (26) in $\dot{V}_z$, which yields:

$$\dot{V}_z = 2z_1\dot{z}_1 + 2z_2\dot{z}_2 = 2z_1(u_1 + \delta_1) + 2z_2(u_2 + \delta_2). \quad (29)$$

When using the proposed control law u_1, u_2 by replacing Eq. (28) in Eq. (29), we obtain:

$$\begin{aligned} \dot{V}_z &= 2z_1(-k_1z_1 - \lambda_1|s|^\rho \operatorname{sgn}(s)z_2 + \delta_1) \\ &\quad + 2z_2(-k_2z_2 + \lambda_2|s|^\rho \operatorname{sgn}(s)z_1 + \delta_2) \\ &= -2k_1z_1^2 - 2k_2z_2^2 - 2\lambda_1|s|^\rho z_1z_2 \operatorname{sgn}(s) \\ &\quad + 2\lambda_2|s|^\rho z_1z_2 \operatorname{sgn}(s) + 2z_1\delta_1 + 2z_2\delta_2. \end{aligned} \quad (30)$$

If we assume $\lambda_1 = \lambda_2$ to cancel the mixed term, this expression is simplified to:

$$\dot{V}_z = -2k_1z_1^2 - 2k_2z_2^2 + 2z_1\delta_1 + 2z_2\delta_2. \quad (31)$$

By considering the disturbances δ_1, δ_2 are bounded such that $|\delta_1| \leq \Delta_1$ and $|\delta_2| \leq \Delta_2$, we can bound $\dot{V}_z$ as follows:

$$\dot{V}_z \leq -2k_1z_1^2 - 2k_2z_2^2 + 2|z_1|\Delta_1 + 2|z_2|\Delta_2. \quad (32)$$

Finally, by using the inequalities $2|z_1|\Delta_1 \leq k_1z_1^2 + \frac{\Delta_1^2}{k_1}$ and $2|z_2|\Delta_2 \leq k_2z_2^2 + \frac{\Delta_2^2}{k_2}$, this expression simplifies to:

$$\dot{V}_z \leq -k_1z_1^2 - k_2z_2^2 + \frac{\Delta_1^2}{k_1} + \frac{\Delta_2^2}{k_2}. \quad (33)$$

Thus, we can verify that $\dot{V}_z \leq 0$ when z_1, z_2 are sufficiently large and $\Delta_1^2/k_1, \Delta_2^2/k_2$ are sufficiently small, ensuring that z_1, z_2 are bounded and will tend to decrease over time. This condition guarantees that V_z does not increase indefinitely, and that z_1, z_2 will converge to sufficiently small values, depending on the applied disturbances and chosen gains k_1, k_2 . Now, to verify the convergence of $s = z_3$, let us analyze V_s time-derivative, and replace Eq. (26) in $\dot{V}_s$, which yields:

$$\dot{V}_s = 2z_3\dot{z}_3 = 2z_3(u_1z_2 - z_1u_2 + \delta_3). \quad (34)$$

When using the proposed control law u_1, u_2 by replacing Eq. (28) in Eq. (34) and considering $s = z_3$, we obtain:

$$\begin{aligned} \dot{V}_s &= 2z_3[(-k_1z_1 - \lambda_1|z_3|^\rho \operatorname{sgn}(z_3)z_2)z_2 \\ &\quad - (-k_2z_2 + \lambda_2|z_3|^\rho \operatorname{sgn}(z_3)z_1)z_1 + \delta_3]. \end{aligned} \quad (35)$$

By observing that $z_3 \operatorname{sgn}(z_3) = |z_3|$ and also assuming that $k_1 = k_2 > 0$, reminding that $\lambda_1 = \lambda_2 > 0$, this simplifies to:

$$\dot{V}_s = -2\lambda_1|z_3|^{\rho+1} + 2z_3\delta_3. \quad (36)$$

By considering the drift-disturbance term $\delta_3 = \zeta + \delta_{o3}$ is bounded such that $|\delta_3| \leq \Delta_3$, we can bound V_s as follows:

$$\dot{V}_s \leq -2\lambda_1|z_3|^{\rho+1} + 2|z_3|\Delta_3. \quad (37)$$

Finally, by verifying that the inequality $2|z_3|\Delta_3 \leq \lambda_1|z_3|^{\rho+1} + \Delta_3^2/(\lambda_1|z_3|^{\rho-1})$ holds, we obtain:

$$\dot{V}_s \leq -\lambda_1|z_3|^{\rho+1} + \frac{\Delta_3^2}{\lambda_1|z_3|^{\rho-1}}. \quad (38)$$

This conclusion shows that, for sufficient small values of $|z_3|$, the negative term $-\lambda_1|z_3|^{\rho+1}$ will dominate, which guarantees that $\dot{V}_s < 0$, driving $z_3 \rightarrow 0$ as $t \rightarrow \infty$.

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